

THE STRONG HYPERKÄHLER SYZ CONJECTURE

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ABSTRACT. We prove the strong hyperkähler SYZ conjecture: every holomorphic line bundle with nonzero nef isotropic first Chern class on a compact irreducible holomorphic symplectic Kähler manifold is semiample.

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1. INTRODUCTION

A compact irreducible holomorphic symplectic Kähler manifold is a simply connected compact Kähler manifold X such that $H^0(X, \Omega_X^2) = \mathbb{C}\sigma$ for a holomorphic two-form σ which is nondegenerate everywhere. Its complex dimension is even, say $2n$. We write $q = q_X$ for the Beauville–Bogomolov–Fujiki quadratic form, and also for its associated symmetric bilinear form. We denote the Kähler cone by $\mathcal{K}_X$. A line bundle L is *nef* if $c_1(L)$ lies in the closure of the Kähler cone, *isotropic* if $q(c_1(L)) = 0$, and *semiample* if some positive tensor power is generated by its global holomorphic sections.

A nonzero nef isotropic class has numerical dimension n , by Fujiki’s relation: its powers have positive intersection with a Kähler class through degree n , and zero intersection in higher degrees. The strong hyperkähler SYZ conjecture asks whether every line bundle with such a class is semiample [43, Conjecture 1.7]. This would turn a class on the boundary of the Kähler cone into a holomorphic map with half-dimensional fibers. The numerical condition by itself supplies neither sections nor control of their base locus.

Theorem 1.1. *Let X be a compact irreducible holomorphic symplectic Kähler manifold of complex dimension $2n$, where $n \geq 1$. If a holomorphic line bundle L satisfies*

$$c_1(L) \neq 0, \quad c_1(L) \in \overline{\mathcal{K}_X}, \quad q_X(c_1(L)) = 0,$$

then L is semiample.

More precisely, there exist a positive integer m , a normal projective variety B , an ample line bundle A on B , and a surjective holomorphic map with connected fibers

$$f : X \longrightarrow B, \quad L^m \simeq f^*A,$$

such that $\dim_{\mathbb{C}} B = n$. Every irreducible component of every fiber has complex dimension n , and σ vanishes on its smooth locus.

Theorem 1.1 resolves the strong hyperkähler SYZ conjecture positively. The conclusion holds on the given manifold X : a positive tensor power of L has no base points anywhere. No restriction on the dimension, deformation type, or second Betti number is imposed.

There is also a consequence for the number of possible smooth manifolds. Here a deformation type means an equivalence class under smooth proper deformations through compact irreducible holomorphic symplectic Kähler manifolds.

Corollary 1.2. *For each integer $n \geq 1$, there are only finitely many smooth deformation types of compact irreducible holomorphic symplectic Kähler manifolds of complex dimension $2n$ with $b_2 \geq 5$.*

This follows by combining Theorem 1.1 with the boundedness of Lagrangian-fibered symplectic varieties proved by Engel–Filipazzi–Greer–Mauri–Svaldi [11]. Section 6 gives the deduction in the smooth category, including the role of the hypothesis $b_2 \geq 5$.

Corollary 1.3 (Projective case). *In Theorem 1.1, assume additionally that X is projective, and let $f : X \rightarrow B$ be the associated connected-fiber map furnished there. Then $B \simeq \mathbb{P}_{\mathbb{C}}^n$. Moreover, there do not exist a prime divisor $D \subset B$, an effective divisor E on X , and an integer $r > 1$ such that $f^*D = rE$.*

This is a consequence of the projective-space base theorem in the companion [36, Theorem 1.1 and the following paragraph on codimension-one multiple fibers]. Its application is given after the proof of Theorem 1.1, in Section 5.6.

1.1. Context. The fibration picture is related to the special Lagrangian formulation of mirror symmetry proposed by Strominger–Yau–Zaslow [41]. In the hyperkähler setting, rotation of complex structure relates special Lagrangian and holomorphic Lagrangian geometry; the original

paper already discusses this relation for K3 surfaces [41, Section 4.1]. Early explicit nef-isotropic formulations appear in Hassett–Tschinkel [18, Remark 3.12] and Sawon [37, Conjecture 4.1]. The former asks for semiample, while the latter relates the existence of a nef square-zero divisor to the existence of an abelian fibration. Matsushita established the half-dimensional and Lagrangian character of fibrations from projective irreducible symplectic manifolds [30, 32]; his equidimensionality theorem also controls every component of every fiber [31, Theorem 1]. Thus, once semiample is known, the geometry of the resulting map fits the expected holomorphic symplectic fibration picture.

Analytic positivity gives another source of progress. Verbitsky proves that a nontrivial nef isotropic bundle with a smooth semipositive Hermitian metric has a nonzero section in some positive power [43, Theorem 1.9]. Höring–Lazić–Lehn obtain nonvanishing or coisotropic-locus alternatives for nef isotropic classes. In complex dimension four, they prove semiample if the class contains a closed positive current whose Lelong numbers all vanish [19, Theorem D and Corollary E]. These results show how additional analytic positivity or geometric information can turn the numerical condition into sections.

The deformation approach has been particularly effective. Soldatenkov and Verbitsky show that a fixed isotropic class is semiample at every nef point of its deformation component as soon as it is semiample at one point [40, Theorem 3.7(v)]. For the known deformation types, earlier constructions and birational results were obtained by Bayer–Macrì [3, Theorem 1.5] and Markman [29, Theorem 1.3 and Remark 1.8] for K3^[n]-type, by Yoshioka [47, Proposition 3.38] for generalized Kummer type, and by Mongardi–Rapagnetta [35, Theorem 7.2] and Mongardi–Onorati [34, Theorem 2.2] for the two O’Grady types. These works combine moduli spaces of sheaves and stability conditions with monodromy and deformation methods. Their conclusions include fibrations on particular moduli spaces, results on the given manifold under nefness, and fibrations on birational models. Soldatenkov–Verbitsky’s deformation theorem implies the strong conjecture for all known hyperkähler deformation classes [40, Section 1.3]. This theorem treats compact Kähler manifolds and preserves the specified class; those two features are essential to our return from a nonprojective rotation to the original pair (X, L) .

The task left by this deformation theorem is to produce one semiample deformation of the *prescribed* class. We do so by constructing a torus and rotating the complex structure. The geometric input after rotation is the theorem of Greb–Lehn–Rollenske: a nonprojective hyperkähler manifold containing a holomorphic Lagrangian torus has a holomorphic Lagrangian fibration [15, Theorem 4.1].

The degeneration construction has a period-theoretic antecedent in Todorov’s work on maximally unipotent degeneration and vanishing torus cycles. His arithmetic-cusp construction relates isotropic classes to degenerations when $b_2 \geq 5$ [42, Section 4]. Here the given class is retained as a labelled vector in the full second-cohomology lattice through family realization and semistable reduction. Starting with the supplied isotropic vector imposes no additional lower bound on the second Betti number.

On the analytic side, Zhang develops the persistence of special Lagrangian tori in Calabi–Yau structures close to a flat cylinder [48, Section 4], using McLean’s deformation operator [33]. Li’s Fermat-family theorem gives special Lagrangian torus fibrations on generic regions whose relative volumes tend to one along a subsequence [24, Theorem 1.1]; [25] surveys the metric SYZ framework. For polarized algebraic maximal degenerations of compact Calabi–Yau manifolds, Blum–Liu’s valuative independence theorem [4, Theorem 1.1] and Li’s comparison theorems [26, Theorems 1.4 and 1.8] give special Lagrangian torus fibrations on open sets whose normalized Calabi–Yau volumes are arbitrarily close to one near the degeneration. The semiample argument needs a local torus together with its coordinate isotopy class, which connects its geometry to the prescribed

monodromy. We obtain that torus by proving convergence to a full smooth flat cylinder in the actual logarithmic coordinates; the final perturbation uses the Zhang–McLean method.

1.2. The proof in four steps. Write $c_1(L) = m_0 e$, where $m_0 > 0$ and e is primitive integral. The proof produces a semiample deformation of e and then returns to X .

- (1) *Retain the prescribed class in monodromy.* Keep e as a labelled integral cohomology class; it need not remain of type $(1, 1)$ during this construction. Choose a polarization on a deformation of X whose marked class h is perpendicular to e . A rational three-dimensional period subspace containing e gives an arithmetic curve with a cusp labelled by $\mathbb{Q}e$. We realize that curve by an algebraic family and track its full second-cohomology monodromy through semistable reduction (Proposition 4.1).
- (2) *Construct a torus in a known isotopy class.* For a Calabi–Yau degeneration of complex dimension d , with $s = -\log |t|$, multiply the polarized Ricci-flat metric by s . Intersection numbers control both traces of this metric relative to logarithmic coordinates. Together with the curvature and projective-map energy estimates, they select noncollapsed centers and produce a metric limit with flat regular part. We identify the full limit with a smooth flat cylinder, including its apparent singularities. In the actual logarithmic coordinates, convergence is smooth on a fixed tube containing an entire angular torus (Lemma 3.4). A small special Lagrangian perturbation retains its coordinate isotopy class (Theorem 2.1). Here special Lagrangian means that the Kähler form vanishes on the torus and the holomorphic volume form has constant phase there.
- (3) *Rotate using cohomology.* For a suitable fiber Y , write its hyperkähler triple as (I, J, K) , with I its original complex structure. The coordinate torus is parameterized consistently around a loop of the family. Its restriction map therefore kills the image of $M - 1$, where M is monodromy on $H^2(Y, \mathbb{R})$. The chosen periods put $[\omega_K]$ in that image. A cohomological rotation criterion (Lemma 5.1) turns the special Lagrangian torus into a holomorphic Lagrangian torus for J . The rotated periods make $Y' = (Y, J)$ nonprojective and force every rational isotropic $(1, 1)$ class to lie on $\mathbb{Q}e$. Thus the Greb–Lehn–Rollenske fibration makes e semiample on Y' .
- (4) *Return to the given manifold.* The markings and the positive sign of e place X and Y' in the same fixed-class deformation locus. The Soldatenkov–Verbitsky theorem then gives semiampleness on X itself. Fujiki’s formula and Matsushita’s equidimensionality theorem give the stated fiber properties.

The cylinder argument extends harmonic functions across the possible singular set to obtain a global derivative bound. Path lifting identifies the metric completion, and bounded holomorphic coframes remove its apparent singularities. In the rotation criterion, cohomological vanishing of one Kähler form suffices: the special Lagrangian period and equality in Wirtinger’s inequality supply the pointwise conclusion.

1.3. Organization and conventions. Sections 2 and 3 prove the torus theorem independently of hyperkähler period theory. Section 4 constructs the prescribed degeneration; Lemma 4.2 separates the underlying polarized family and free period quotient from the cusp calculation. Section 5 rotates the torus and proves Theorem 1.1; Section 6 deduces Corollary 1.2.

Unspecified cohomology coefficients are real. We normalize q integrally; its signature is $(3, b_2 - 3)$, and the same symbol denotes its associated symmetric bilinear form. Fujiki’s relation is

$$(1.1) \quad \int_X a^{2n} = c_X q(a)^n, \quad c_X > 0.$$

The external inputs are stated where used: the Calabi–Yau theorem, hyperkähler period and Kähler-cone theory, arithmetic period-map algebraicity, semistable reduction, noncollapsed Einstein-limit and harmonic-function theory, and the fibration, deformation, and boundedness theorems cited above.

2. METRIC ESTIMATES AT THE TORUS SCALE

This section and the next construct a special Lagrangian torus near a maximal crossing of a polarized semistable degeneration. The coordinate isotopy class is essential: an angular coordinate torus is parameterized consistently around the puncture, so restriction to it is invariant under monodromy. Section 5 uses this identity in degree two. The construction applies to Calabi–Yau manifolds independently of the hyperkähler setting. We use d for the complex dimension and $N = 2d$ for the real dimension.

Theorem 2.1 (A special Lagrangian torus near a maximal crossing). *Let $\pi : \mathcal{Y} \rightarrow \Delta$ be the restriction to a disk of a projective algebraic family over a smooth curve. Assume that $\mathcal{Y}$ is smooth, $\mathcal{Y}_0$ is a reduced simple normal crossings divisor, and the fibers Y_t for $t \neq 0$ are smooth connected projective manifolds of complex dimension $d \geq 2$ with trivial canonical bundle. Let a_t be polarizing classes given by a line bundle on $\mathcal{Y}$; this line bundle need not be relatively ample at 0. Let η_t be nowhere vanishing holomorphic d -forms which extend as a section of the relative logarithmic canonical line bundle. Put $s = -\log |t|$, and suppose that, for constants $c, C > 0$,*

$$(2.1) \quad cs^d \leq I_t := \int_{Y_t} |\eta_t|^2 \leq Cs^d, \quad 0 < |t| \ll 1.$$

Suppose, moreover, that at some point $o \in \mathcal{Y}_0$ exactly $d+1$ components meet and that the logarithmic coefficient of η at o is nonzero. Thus in a coordinate neighborhood of o ,

$$(2.2) \quad z_0 z_1 \cdots z_d = t, \quad \eta_t = f(z) \frac{dz_1}{z_1} \wedge \cdots \wedge \frac{dz_d}{z_d}, \quad f(o) \neq 0.$$

Equip Y_t with the Ricci-flat Kähler metric in a_t . For every sufficiently small nonzero t , the fiber Y_t contains a smooth embedded special Lagrangian real d -torus isotopic in Y_t to an angular coordinate torus in (2.2). Here special Lagrangian means that the Kähler form restricts to zero and that η_t has constant phase on the oriented torus.

We begin with an arbitrary sequence of nonzero parameters tending to zero. The compactness argument will produce a subsequence on which every sufficiently late fiber contains the required torus. At the end of Section 3, a sequence of counterexamples will rule out failure anywhere in a sufficiently small punctured disk.

This section chooses centers with noncollapsed unit balls and controls the metric and logarithmic coordinates on every ball about them. Section 3 identifies the pointed limit as a smooth flat cylinder and perturbs its angular torus. All constants below are independent of t in a sufficiently small punctured disk, unless stated otherwise. Every extraction is from the arbitrary initial sequence.

2.1. Coordinate volume and mixed intersections. Let $g_t^{(0)}$ and $\omega_t^{(0)}$ be the Ricci-flat metric and Kähler form supplied by the Calabi–Yau theorem [46], with $[\omega_t^{(0)}] = a_t$. The useful scale is

$$(2.3) \quad g_t = sg_t^{(0)}, \quad \omega_t = s\omega_t^{(0)}, \quad \mu_t = dV_t = \frac{\omega_t^d}{d!}, \quad \mu_t(Y_t) = c_a s^d, \quad c_a = \frac{1}{d!} \int_{Y_t} a_t^d > 0.$$

The last constant is independent of t by topological local triviality over the punctured disk. Scaling does not change either of the special Lagrangian conditions.

Take $z_1, \dots, z_d$ to be algebraic local equations of the corresponding components. They extend to rational functions on the algebraic total space. The remaining unit in the local equation of π can be absorbed into z_0 ; no rationality of this last coordinate will be used. Shrink the analytic coordinate neighborhood so that $|z_i| < r_* < 1$ there. On its smooth fibers set

$$x_i = \log |z_i|, \quad y_i = \arg z_i \in \mathbb{R}/(2\pi\mathbb{Z}), \quad \gamma = \sum_{i=1}^d (dx_i^2 + dy_i^2), \quad \omega_\gamma = \sum_{i=1}^d dx_i \wedge dy_i.$$

Thus $x_i + iy_i$ are local holomorphic logarithms, and γ is the standard flat metric on $\mathbb{R}^d \times (\mathbb{R}/2\pi\mathbb{Z})^d$. Its angular circles have fixed length; we will compare g_t with this metric. Let

$$\Sigma = \left\{ b \in \mathbb{R}^d : b_i < 0 \ (1 \leq i \leq d), \quad \sum_i b_i > -1 \right\}.$$

For a relatively compact subset $B \Subset \Sigma$, write

$$\mathcal{T}_t(B) = \{x/s \in B, \quad y \in (\mathbb{R}/2\pi\mathbb{Z})^d\}.$$

For small t this tube is contained in the coordinate neighborhood. Indeed all $|z_i|$, including $|z_0| = \exp(-s - \sum_i x_i)$, are bounded by $e^{-\delta_B s}$ for some $\delta_B > 0$. In particular, these tubes converge uniformly to o in the total space.

Use the convention $|\eta_t|^2 = 2^{-d} i^{d^2} \eta_t \wedge \overline{\eta}_t$, so that (2.2) gives $|\eta_t|^2 = |f(z)|^2 dV_\gamma$ in the chart. Ricci-flatness implies that its volume form is a constant multiple of $|\eta_t|^2$; integration determines that constant. Hence

$$(2.4) \quad dV_t = \frac{c_a s^d}{I_t} |\eta_t|^2, \quad \frac{dV_t}{dV_\gamma} = \frac{c_a s^d}{I_t} |f(z)|^2 \longrightarrow \vartheta > 0 \quad \text{uniformly on } \mathcal{T}_t(B)$$

after a subsequence. To obtain the convergence, first extract a limit of $c_a s^d / I_t$ using (2.1), and then use $f(z) \rightarrow f(o) \neq 0$. In particular, the two volume measures are uniformly comparable on each fixed interior tube.

Lemma 2.2 (Mixed-intersection estimates). *For every fixed $B \Subset \Sigma$ and $0 \leq k \leq d$,*

$$(2.5) \quad \int_{\mathcal{T}_t(B)} \omega_\gamma^k \wedge \omega_t^{d-k} \leq C_B s^d.$$

Consequently, with traces taken for real Riemannian metrics,

$$(2.6) \quad \int_{\mathcal{T}_t(B)} \text{tr}_\gamma g_t dV_\gamma \leq C_B s^d, \quad \int_{\mathcal{T}_t(B)} \text{tr}_{g_t} \gamma dV_t \leq C_B s^d.$$

Proof. Resolve the finitely many rational maps $z_i : \mathcal{Y} \dashrightarrow \mathbb{P}^1$ simultaneously. We obtain a projective modification $\varpi : \widehat{\mathcal{Y}} \rightarrow \mathcal{Y}$ and morphisms $f_i : \widehat{\mathcal{Y}} \rightarrow \mathbb{P}^1$, with ϖ an isomorphism over their common regular locus, in particular over the coordinate neighborhood being used. After shrinking the punctured disk, its nearby fibers $\widehat{Y}_t$ are smooth. This follows from generic smoothness for the fixed algebraic model; no resolution depending on s is required.

Fix a Fubini–Study form ω_{FS} on $\mathbb{P}^1$. For $A_b([u : v]) = [e^b u : v]$, define the smooth closed semipositive form

$$\alpha_{s,t} = \frac{1}{s} \int_0^s \sum_{i=1}^d f_i^* A_b^* \omega_{\text{FS}} db \quad \text{on } \widehat{Y}_t.$$

Its cohomology class is the restriction of $\sum_i f_i^* [\omega_{\text{FS}}]$, independently of s , since each A_b is homotopic to the identity. Thus every intersection

$$(2.7) \quad \int_{\widehat{Y}_t} \alpha_{s,t}^k \wedge (\varpi^* \omega_t^{(0)})^{d-k}$$

is a fixed finite number. Both classes involved extend over the fixed model, so this number is independent of t as well.

In a logarithmic coordinate $w_i = x_i + iy_i$, the i th summand before averaging is a fixed positive constant times

$$\frac{e^{2(b+x_i)}}{(1 + e^{2(b+x_i)})^2} dx_i \wedge dy_i.$$

For $x/s \in B$, both $-x_i$ and $s + x_i$ are at least $\delta_B s$. Consequently the interval $[-x_i - 1, -x_i + 1]$ lies in $[0, s]$ for large s . On this interval the displayed scalar factor is bounded below by a positive constant. Summing over i proves, as Hermitian forms on the tube,

$$\alpha_{s,t} \geq \frac{c_B}{s} \omega_\gamma.$$

The pullback $\varpi^* \omega_t^{(0)}$ is semipositive, including on the exceptional divisors. Positivity of wedge products therefore gives

$$\begin{aligned} \int_{\mathcal{T}_t(B)} \omega_\gamma^k \wedge \omega_t^{d-k} &\leq C_B s^k s^{d-k} \int_{\widehat{Y}_t} \alpha_{s,t}^k \wedge (\varpi^* \omega_t^{(0)})^{d-k} \\ &\leq C'_B s^d. \end{aligned}$$

This proves (2.5). Finally, for Hermitian metrics h, k with Kähler forms ω_h, ω_k , the real trace identity is

$$(\mathrm{tr}_h k) dV_h = \frac{2}{(d-1)!} \omega_k \wedge \omega_h^{d-1}.$$

Use $k = 1$ and $k = d - 1$ in (2.5) to obtain the two assertions in (2.6). $\square$

2.2. Global energies and many noncollapsed centers. The two trace bounds have different roles. The bound for $\mathrm{tr}_{g_t} \gamma$ controls logarithmic functions on intrinsic balls; the bound for $\mathrm{tr}_\gamma g_t$ supplies many noncollapsed centers. We also record two energies whose averages tend to zero.

Choose concentric open Euclidean balls $B_0 \Subset B_1 \Subset B_2 \Subset \Sigma$, and a smooth function $0 \leq \rho \leq 1$ supported compactly in B_2 and identically one on B_1 . We will select centers in $\mathcal{T}_t(B_0)$. Define functions on the entire compact fiber by extending the following by zero:

$$(2.8) \quad U_0 = s\rho(x/s), \quad U_j = x_j \rho(x/s) \quad (1 \leq j \leq d), \quad H_t = \sum_{j=0}^d |\nabla U_j|_{g_t}^2.$$

These extensions are smooth because their support is compactly contained in the chart for each fixed t . On this support x_j/s is bounded, and

$$dU_0 = \sum_i \rho_i(x/s) dx_i, \quad dU_j = \rho(x/s) dx_j + \frac{x_j}{s} \sum_i \rho_i(x/s) dx_i.$$

It follows that $H_t \leq C \mathrm{tr}_{g_t} \gamma$ on the support and that

$$(2.9) \quad \frac{1}{\mu_t(Y_t)} \int_{Y_t} H_t dV_t \leq C_H.$$

Choose a fixed projective embedding of the family over a neighborhood of the disk, written $(\Phi, \pi) : \mathcal{Y} \hookrightarrow \mathbb{P}^M \times \Delta$. Give $\mathbb{P}^M$ a fixed Fubini–Study metric and set

$$K_t = |\mathrm{Rm}_{g_t}|^2 + |d\Phi_t|_{g_t}^2.$$

The Ricci-flat Kähler Chern–Weil identity expresses $\int_{Y_t} |\mathrm{Rm}_{g_t^{(0)}}|^2 dV_{g_t^{(0)}}$ as a universal dimension-dependent positive constant times $\int_{Y_t} c_2(Y_t) a_t^{d-2}$. This is independent of t . The energy of the holomorphic map Φ_t in the unscaled metric is likewise a fixed multiple of $\int_{Y_t} \Phi_t^* [\omega_{\mathrm{FS}}] a_t^{d-1}$. Under $g_t = s g_t^{(0)}$, squared curvature scales by s^{-2} , map-energy density by s^{-1} , and volume by s^d .

Therefore

$$\begin{aligned}
 (2.10) \quad & \int_{Y_t} |\operatorname{Rm}_{g_t}|^2 dV_t \leq C s^{d-2}, \\
 & \int_{Y_t} |d\Phi_t|_{g_t}^2 dV_t \leq C s^{d-1}, \\
 & m_t := \frac{1}{\mu_t(Y_t)} \int_{Y_t} K_t dV_t \leq C(s^{-2} + s^{-1}) \longrightarrow 0.
 \end{aligned}$$

This also covers $d = 2$, when the curvature integral in the unscaled metric is purely topological.

Lemma 2.3 (A positive fraction of noncollapsed centers). *There exist constants $\delta, v > 0$ and measurable sets $E_t \subset \mathcal{T}_t(B_0)$ such that*

$$(2.11) \quad \mu_t(E_t) \geq \delta \mu_t(Y_t), \quad \mu_t(B_{g_t}(q, 1)) \geq v \quad (q \in E_t).$$

All balls here and below use the intrinsic distance of the complete manifold (Y_t, g_t) .

Proof. Fix a rectangular angle chart of positive size in $(\mathbb{R}/2\pi\mathbb{Z})^d$. The region sB_0 contains at least $c_0 s^d$ disjoint translates of a fixed small d -dimensional cube. Taking their products with one fixed angle cube gives at least $c_0 s^d$ disjoint convex coordinate boxes Q , all of the same Euclidean size, inside $\mathcal{T}_t(B_0)$. By the first estimate of (2.6), at least half of these boxes satisfy

$$(2.12) \quad \int_Q \operatorname{tr}_\gamma g_t dV_\gamma \leq C_0$$

for a constant C_0 independent of t . By (2.4), every such box has $m_0 \leq \mu_t(Q) \leq m_1$ for fixed positive m_0, m_1 .

For $a, b \in Q$, the straight coordinate segment lies in Q , and its length bounds the intrinsic distance from above. Consequently

$$d_{g_t}(a, b) \leq |a - b|_\gamma \int_0^1 \sqrt{\operatorname{tr}_\gamma g_t((1-r)a + rb)} dr.$$

Write $h = \sqrt{\operatorname{tr}_\gamma g_t}$. For $0 \leq r \leq 1/2$, first integrate in a and change variables $z = (1-r)a + rb$. Its Jacobian is $(1-r)^N \geq 2^{-N}$ and z stays in Q . Thus

$$\int_Q \int_Q h((1-r)a + rb) da db \leq 2^N \operatorname{vol}_\gamma(Q) \int_Q h dV_\gamma.$$

For $1/2 \leq r \leq 1$, use b first and the Jacobian r^N instead. The Euclidean diameter and volume of Q are fixed. Cauchy–Schwarz and (2.12) now bound the double integral of $d_{g_t}(a, b)$ uniformly. Comparing measures gives a uniform constant C_1 such that

$$\frac{1}{\mu_t(Q)^2} \int_Q \int_Q d_{g_t}(a, b) dV_t(a) dV_t(b) \leq C_1.$$

The set E_Q of $a \in Q$ for which $\mu_t(Q)^{-1} \int_Q d_{g_t}(a, b) dV_t(b) \leq 2C_1$ has measure at least $\mu_t(Q)/2$. For each such a , a second application of Markov's inequality gives

$$\mu_t(B_{g_t}(a, R_0)) \geq \mu_t(Q)/2 \geq m_0/2, \quad R_0 = \max\{1, 4C_1\}.$$

Since $\operatorname{Ric}(g_t) = 0$, Bishop–Gromov comparison yields $\mu_t(B_{g_t}(a, 1)) \geq R_0^{-N} m_0/2$. Finally take the union of E_Q over the disjoint good boxes. There are at least $(c_0/2)s^d$ of them and each contributes measure at least $m_0/2$. In view of (2.3), this union has a fixed positive fraction of the total volume, as required. $\square$

2.3. One center with estimates at every radius. We now choose one of the noncollapsed centers so that the average cutoff energy stays bounded, and the average curvature and embedding energies tend to zero, on every ball about it. The maximal function makes this simultaneous choice possible.

For a nonnegative integrable function h on Y_t , let

$$\mathcal{M}_t h(q) = \sup_{r>0} \frac{1}{\mu_t(B_{g_t}(q, r))} \int_{B_{g_t}(q, r)} h \, dV_t$$

be the centered maximal function. Bishop–Gromov gives doubling with constant 2^N . The usual disjoint-ball covering argument therefore has a constant D_N independent of t and gives

$$(2.13) \quad \mu_t\{\mathcal{M}_t h > \lambda\} \leq \frac{D_N}{\lambda} \int_{Y_t} h \, dV_t.$$

For completeness, take at each point of the set on the left a ball whose average exceeds λ , select disjoint balls whose fivefold enlargements cover a compact subset of that set, and use $\mu_t(5B) \leq 5^N \mu_t(B)$. Sum the integrals on the disjoint balls and exhaust the set by compact subsets. Radii larger than the diameter simply give the whole manifold, so no diameter bound is needed.

Choose A so large that $D_N C_H / A < \delta/3$. Set $\epsilon_t = \sqrt{m_t} + s^{-1}$, which is positive and tends to zero. The estimate (2.13), first for H_t and then for K_t , shows that

$$\mu_t\{\mathcal{M}_t H_t > A\} < \frac{\delta}{3} \mu_t(Y_t), \quad \mu_t\{\mathcal{M}_t K_t > \epsilon_t\} \leq D_N \sqrt{m_t} \mu_t(Y_t).$$

For small t the latter quantity is also less than $\delta \mu_t(Y_t)/3$. These two exceptional sets cannot cover E_t . We can therefore select $p_t \in E_t$ with

$$(2.14) \quad \begin{aligned} \mu_t(B_{g_t}(p_t, 1)) &\geq v, \\ \frac{1}{\mu_t(B_{g_t}(p_t, R))} \int_{B_{g_t}(p_t, R)} H_t \, dV_t &\leq A, \\ \frac{1}{\mu_t(B_{g_t}(p_t, R))} \int_{B_{g_t}(p_t, R)} K_t \, dV_t &\leq \epsilon_t \quad \text{for every } R > 0. \end{aligned}$$

After a further subsequence,

$$b_t := x(p_t)/s \longrightarrow b_* \in \overline{B_0} \subset B_1.$$

In particular $\rho(b_t) = 1$, $U_0(p_t) = s$, and $U_j(p_t) = x_j(p_t)$.

We next turn average gradient control into an estimate centered at the actual value of the function at p_t . The ball Poincaré inequality for nonnegative Ricci curvature, a consequence of the segment inequality [6], states that

$$\frac{1}{\mu_t(B(q, r))} \int_{B(q, r)} |U - U_{B(q, r)}|^2 \, dV_t \leq C_N r^2 \frac{1}{\mu_t(B(q, 2r))} \int_{B(q, 2r)} |\nabla U|^2 \, dV_t.$$

Here $U_{B(q, r)}$ denotes the mean and doubling absorbs any equivalent choice of enlarged radius. Applied at p_t to U_j , this gives an upper bound $C A r^2$. Writing $a_r = (U_j)_{B(p_t, r)}$ and comparing the smaller ball with the larger one by doubling, we also obtain

$$\begin{aligned} |a_{r/2} - a_r|^2 &\leq \frac{1}{\mu_t(B(p_t, r/2))} \int_{B(p_t, r/2)} |U_j - a_r|^2 \, dV_t \\ &\leq C A r^2. \end{aligned}$$

Since U_j is smooth, $a_{2^{-k}R} \rightarrow U_j(p_t)$ as $k \rightarrow \infty$. Summing the preceding bounds over dyadic radii yields $|a_R - U_j(p_t)| \leq C \sqrt{A} R$. Combining this with Poincaré proves

$$(2.15) \quad \frac{1}{\mu_t(B_{g_t}(p_t, R))} \int_{B_{g_t}(p_t, R)} |U_j - U_j(p_t)|^2 \, dV_t \leq C R^2 \quad (0 \leq j \leq d, \, R > 0).$$

The cutoff functions now locate each fixed intrinsic ball in the crossing chart. Divide (2.15) by s^2 and use $b_t \rightarrow b_*$. On every fixed-radius ball, $U_0/s \rightarrow 1$ and $U_j/s \rightarrow (b_*)_j$ in L^2 . Where $U_0/s > 1/2$, the point lies in the coordinate tube and

$$\frac{x_j}{s} = \frac{U_j/s}{U_0/s}.$$

Chebyshev's inequality consequently gives, for every $R, \eta > 0$,

$$(2.16) \quad \mu_t(B_{g_t}(p_t, R) \setminus \{q \text{ in the chart} : |x(q)/s - b_*| < \eta\}) \rightarrow 0.$$

The assertion concerns absolute volume, because Bishop–Gromov gives the uniform bound $\mu_t(B_{g_t}(p_t, R)) \leq \omega_N R^N$, where ω_N is the volume of the Euclidean unit ball.

2.4. The regular part of the pointed limit. Noncollapse permits us to pass to a limit. Vanishing curvature energy will make its regular set flat; vanishing embedding energy will place each regular compact set inside the chosen crossing chart.

The noncollapse in (2.14) and $\text{Ric}(g_t) = 0$ give, after extraction, a pointed measured Gromov–Hausdorff limit

$$(Y_t, g_t, p_t, \mu_t) \rightarrow (Z, d_Z, p, \mu).$$

We use the following precise consequences of noncollapsed Ricci limit theory and Einstein regularity [7, 10, 9]. The space Z is complete and proper; μ is its N -dimensional Riemannian volume on its open dense regular set G . Regular points are the points with a Euclidean tangent cone. Compact subsets of G admit smooth convergence identifications with regions in Y_t , under which the metrics converge in C^∞ . The closed singular set $S = Z \setminus G$ satisfies

$$(2.17) \quad \dim_{\text{H}} S \leq N - 4, \quad \mu(B_Z(q, r)) \leq \omega_N r^N.$$

The codimension assertion is Theorem 1.4 of [9]; the smooth convergence follows from Einstein ϵ -regularity [9, Theorem 2.11] and elliptic bootstrapping. Complex structures also converge smoothly on G : they are parallel orthogonal tensors for the converging metrics. Their limit I_∞ is parallel and integrable, so the limiting metric g is Kähler on G .

Every compact smooth region lies in the image of a fixed-radius ball about p_t for all sufficiently small t along the sequence. By (2.14) and the volume upper bound, its curvature energy is at most $\epsilon_t \omega_N R^N$, which tends to zero. Smooth metric convergence therefore implies

$$(2.18) \quad \text{Rm}_g = 0 \quad \text{on } G.$$

It remains to put these smooth regions in the coordinate chart about o . We use the following small-energy argument for the maps Φ_t .

Lemma 2.4 (Vanishing energy on a smooth domain). *Let h_ν be metrics converging smoothly on a fixed smooth domain D of real dimension N , and let $\psi_\nu : (D, h_\nu) \rightarrow P$ be smooth harmonic maps to a fixed compact Riemannian manifold. If their energies tend to zero on every relatively compact subdomain, then $\sup_K |d\psi_\nu| \rightarrow 0$ for every $K \Subset D$. After extraction they converge smoothly to a constant on each connected relatively compact domain.*

Proof. Smooth harmonic maps are stationary, which is the property used in the energy monotonicity argument below. Fix nested relatively compact domains around K . Their metrics have uniform smooth bounds, and geodesic balls with centers in an intermediate domain have a uniformly positive admissible radius r_0 . For a smooth harmonic map the stress tensor $|d\psi|^2 h - 2\psi^* g_P$ has zero divergence. Testing this identity against radial vector fields and using $\nabla^2(r^2/2) = h + O(r^2)$ in these geodesic balls gives

$$\Theta'_\nu(q, r) \geq -Cr\Theta_\nu(q, r), \quad \Theta_\nu(q, r) = r^{2-N} \int_{B_{h_\nu}(q, r)} |d\psi_\nu|^2 dV_{h_\nu}.$$

The nonnegative radial derivative term has been discarded here. Integration gives the uniform monotonicity consequence

$$(2.19) \quad \Theta_\nu(q, r) \leq C\Theta_\nu(q, r_0) \leq Cr_0^{2-N}E_\nu \longrightarrow 0 \quad (0 < r \leq r_0),$$

where E_ν is the energy on a fixed larger domain. This is the stationary harmonic-map monotonicity argument; see also [27].

If derivatives were unbounded on K , maximize $\text{dist}_{h_\nu}(q, \partial D')|d\psi_\nu(q)|$ on a fixed intermediate domain D' . At maximizing points q_ν , put $L_\nu = |d\psi_\nu(q_\nu)|$. Then $L_\nu \rightarrow \infty$ and the product just maximized tends to infinity. On the ball of radius L_ν^{-1} about q_ν , point selection gives $|d\psi_\nu| \leq 2L_\nu$ for large ν . Rescale the domain metric by L_ν^2 . The rescaled maps v_ν on unit balls satisfy $|dv_\nu(0)| = 1$ and $|dv_\nu| \leq 2$, while the rescaled domain metrics converge smoothly to a Euclidean metric. The harmonic-map Bochner inequality and these derivative bounds imply

$$\Delta|dv_\nu|^2 \geq -C|dv_\nu|^2.$$

The local mean-value inequality, with uniform constants on these smooth balls, now gives

$$1 \leq C \int_{B(0,1)} |dv_\nu|^2 = CL_\nu^{N-2} \int_{B_{h_\nu}(q_\nu, L_\nu^{-1})} |d\psi_\nu|^2 dV_{h_\nu} = C\Theta_\nu(q_\nu, L_\nu^{-1}) \longrightarrow 0,$$

a contradiction. Thus the derivatives are uniformly bounded on interior domains. Apply the same Bochner and mean-value argument on fixed smaller balls: it gives $\sup_K |d\psi_\nu|^2 \leq CE_\nu \rightarrow 0$. Compactness of P and interior elliptic estimates for the smooth harmonic-map equation then give smooth subsequential convergence; the limit has zero derivative and is constant on connected domains. $\square$

The maps Φ_t are holomorphic between Kähler manifolds and hence harmonic. On a compact subdomain of G , pull them back using the smooth convergence identifications. Their energies tend to zero by (2.14), so Lemma 2.4 applies. Every local constant limit is $\Phi(o)$. Indeed a fixed smaller regular ball has positive limiting volume, whereas the exceptional set in (2.16) has volume tending to zero. That ball therefore contains points with x/s in a fixed compact subset of Σ . Such points converge to o in the total space, and their images converge to $\Phi(o)$. Local uniform convergence of the maps identifies the constant on each regular ball separately, regardless of whether G is connected. It follows, by a finite cover, that

$$(2.20) \quad \Phi_t \longrightarrow \Phi(o) \quad \text{uniformly on every compact subset of } G.$$

Since (Φ, π) is an embedding, there is a neighborhood of $(\Phi(o), 0)$ whose inverse image lies in the coordinate neighborhood (2.2). Thus the corresponding regions of Y_t are eventually in that chart. On such a region all $|z_i| < r_* < 1$, and $\sum_{i=0}^d \log |z_i| = -s$. For $1 \leq j \leq d$ this gives

$$-s - d \log r_* \leq x_j \leq \log r_*.$$

In particular x_j/s is uniformly bounded there. Each x_j is pluriharmonic, being locally the real part of $\log z_j$, and is therefore harmonic for g_t . Its boundedness and (2.16) give local L^2 convergence of x_j/s to $(b_*)_j$ on the smooth regions. Interior harmonic estimates for the smoothly converging metrics improve this to

$$(2.21) \quad \frac{x_j}{s} \longrightarrow (b_*)_j \quad \text{in } C_{\text{loc}}^\infty(G), \quad 1 \leq j \leq d.$$

As $b_* \in B_1$ and $\rho = 1$ there, every compact subset of G has a neighborhood on whose corresponding regions $\rho(x/s) = 1$ for all sufficiently small t . Hence on these regions $U_j = x_j$ for $j \geq 1$. The all-radius energy bound (2.14) and the centered oscillation estimate (2.15) therefore apply to the actual logarithmic coordinates. In the next section they give a global coordinate map on the limit, including its possible singular points.

3. A FLAT CYLINDER AND A SPECIAL LAGRANGIAN TORUS

We continue the proof of Theorem 2.1. All limits are taken along the subsequence selected in the preceding section. We write μ for the limit Riemannian volume and

$$H = \mathbb{C}^d / (2\pi i \mathbb{Z}^d), \quad N = 2d.$$

The standard flat metric on H is again denoted by γ . Our goal is to identify Z with this cylinder, obtain smooth convergence in the actual logarithmic coordinates, and perturb an angular torus. The possible singular set $S = Z \setminus G$ must be handled first: flatness on G alone does not give convergence over an entire torus.

3.1. The coordinate map and its derivative bounds. On every compact subset of the regular set G the actual logarithmic chart is defined for all sufficiently small t , and $\rho = 1$ there. The functions

$$x_j - U_j(p_t), \quad j = 1, \dots, d,$$

are harmonic on these subsets. Their local L^2 bounds, from (2.15), and interior elliptic estimates give smooth subsequential limits u_j . The angular coordinates have values in the compact circles $\mathbb{R}/(2\pi\mathbb{Z})$ and satisfy $d \arg z_j = -dx_j \circ I_t$, where I_t is the fiber complex structure. A diagonal subsequence therefore also gives smooth circle-valued limits ϑ_j . Thus

$$(3.1) \quad F = (u_1 + i\vartheta_1, \dots, u_d + i\vartheta_d) : G \longrightarrow H$$

is holomorphic. Diagonal extraction here and below is over a countable exhaustion of the smooth manifold G .

The coordinate density in (2.4) converges to a positive constant. Consequently the absolute real Jacobian of F , computed using the limit metric and γ , is the constant $J_0 = \vartheta^{-1} > 0$. In particular F is a local biholomorphism. It is also injective. Indeed, if two distinct regular points had the same image, choose disjoint small regular neighborhoods of these points on which F is invertible. Smooth convergence and the inverse function theorem imply that their approximating logarithmic charts both contain one fixed smaller neighborhood of the common image. This contradicts the injectivity of the logarithmic chart on each nearby fiber.

For every $R > 0$, exhaustion of $B(p, R) \cap G$ by regular compact subsets, smooth convergence, and the estimates at p_t give, with a fixed constant C ,

$$(3.2) \quad \int_{B(p, R) \cap G} |u_j|^2 d\mu \leq CR^2 \mu(B(p, 2R)),$$

$$(3.3) \quad \int_{B(p, R) \cap G} |du_j|^2 d\mu \leq C\mu(B(p, 2R)).$$

Enlarging the radii slightly before taking limits avoids any issue with boundaries of balls.

To extend these functions across S , we use the Sobolev structure on the whole noncollapsed Ricci limit. Pointed measured Gromov–Hausdorff convergence implies pointed measured Gromov convergence by [14, Proposition 3.30]. For finite $N > 2$, Lott–Villani’s [28, Corollary E.44] preserves nonnegative N -Ricci curvature under pointed measured Gromov–Hausdorff convergence; in particular, the limit satisfies $\text{CD}(0, N)$. Here $N = 2d \geq 4$. The finite-dimensional curvature-dimension condition in the pointed setting is also treated in [14, Section 4.2], while [14, Theorem 7.2] preserves quadratic Cheeger energy. These statements allow measures finite on bounded sets, without a finite-total-volume assumption. The limit measure has full support: every open set contains a regular ball of positive Riemannian volume. Together with [12, Theorem 3.17(i)], these stability results give the $\text{RCD}(0, N)$ structure used below. At zero curvature the reduced and ordinary curvature-dimension conditions coincide.

On a sufficiently small regular ball, short minimizing Z -segments stay in a larger regular ball. Ambient and Riemannian distances therefore agree locally. Restrictions of Lipschitz approximants

and classical weak lower semicontinuity give the lower bound for the relaxed Sobolev energy; smooth compactly supported approximants, extended by zero, give the upper bound. Locality and exhaustion therefore identify this energy on G with ordinary Riemannian Sobolev energy.

Lemma 3.1. *Each u_j extends to a weakly harmonic function in $W_{\text{loc}}^{1,2}(Z)$. Moreover $|du_j| \leq C$ on G , with a constant independent of the point. Both dF and its local inverse derivatives are uniformly bounded.*

Proof. Sobolev extension. Recall that $\dim_{\mathbb{H}} S \leq N-4$ and $\mu(B(q, r)) \leq \omega_N r^N$. On a fixed bounded subdomain, cover its compact singular portion by balls of radii r_i with $\sum_i r_i^{N-2}$ arbitrarily small, and take a finite subcover. A Lipschitz cutoff on a doubled covering ball has squared gradient integral at most $C r_i^{N-2}$. Taking minima of the complementary cutoffs, and exhausting bounded subdomains, gives cutoffs χ_k which vanish near S , converge to one off S , satisfy $0 \leq \chi_k \leq 1$, and have $\int |d\chi_k|^2 \rightarrow 0$. Their exceptional supports can also be chosen to have measure tending to zero. All these assertions are local; one inserts a fixed exterior cutoff when necessary.

Let $T_A(u_j) = \max(-A, \min(A, u_j))$. For fixed A , the products $\chi_k T_A(u_j)$ are Sobolev functions across S , after extension by zero there. To justify this before any global path-distance assertion, insert a fixed compactly supported exterior cutoff. For each fixed k the product has compact support inside G . The local agreement of distances just noted, applied on a finite regular cover of the support, shows that the product is locally ambient Lipschitz; boundedness controls pairs separated by a fixed positive distance. It is therefore an ambient Sobolev function, with its ordinary Riemannian energy on G . The extra derivative term has norm at most $A \|d\chi_k\|_2$. Weak compactness and lower semicontinuity give a Sobolev extension of $T_A(u_j)$ with the energy bound inherited from G . Letting $A \rightarrow \infty$ and using (3.2)–(3.3) gives the claimed Sobolev extension of u_j . For a compactly supported Lipschitz function φ , harmonicity on G gives

$$0 = \int_G \chi_k \langle du_j, d\varphi \rangle d\mu + \int_G \varphi \langle du_j, d\chi_k \rangle d\mu.$$

The second integral tends to zero by Cauchy–Schwarz, using (3.3). The first converges to the integral over Z , since S has zero measure. This proves weak harmonicity.

Global derivative bounds. We use the local harmonic gradient estimate on noncollapsed Ricci limits, or equivalently its $\text{RCD}(0, N)$ formulation [20, Theorem 1.2 and Theorem 4.1(a)]:

$$(3.4) \quad \sup_{B(q, R/2)} |du| \leq \frac{C(N)}{R} \left(\frac{1}{\mu(B(q, 2R))} \int_{B(q, 2R)} |u|^2 d\mu \right)^{1/2}.$$

For this signed-function version, the mean-value estimate applied to u^2 bounds $\sup_{B(q, R)} |u|$ by a constant times the normalized L^2 norm on $B(q, 2R)$. Add twice this supremum and apply the positive-harmonic gradient estimate on the smaller ball; if the supremum vanishes, the conclusion is immediate. Fix q and take $R \geq 2d_Z(p, q)$. Then $B(q, 2R) \subset B(p, 3R)$ and $B(p, 6R) \subset B(q, 7R)$, so doubling compares $\mu(B(q, 2R))$ with $\mu(B(p, 6R))$. Equation (3.2) bounds the right side of (3.4) by a constant independent of q and R . Thus $|du_j| \leq C$ on G .

Holomorphicity bounds the angular derivatives as well, so that $|dF| \leq A$ for a fixed A . Since the product of its N real singular values is $J_0 > 0$, its smallest singular value is at least J_0/A^{N-1} . This gives the inverse derivative bound. $\square$

3.2. The metric completion is the full cylinder. We have an injective local biholomorphism on G with uniformly bounded derivative and inverse derivative. To extend this identification over Z , we first compare the intrinsic distance on G with the limit distance, and then continue the inverse along paths in the cylinder.

Lemma 3.2. *The intrinsic path distance on G equals the restriction of d_Z . In particular F extends to a Lipschitz map $F : Z \rightarrow H$.*

Proof. Fix $a, b \in G$ and small balls A, B about them whose closures are contained in G . All minimizing segments between these balls lie in one compact ball. Cover its singular portion by sufficiently small balls $B(q_i, r_i)$, disjoint from $A \cup B$ even after doubling, such that $\sum_i r_i^{N-1}$ is as small as desired. Put

$$f = \sum_i r_i^{-1} \mathbf{1}_{B(q_i, 2r_i)}.$$

The volume upper bound gives $\int f d\mu \leq C \sum_i r_i^{N-1}$. The segment inequality for nonnegative integrable functions on Ricci limits [8, Theorem 2.15] bounds the average, over endpoints in A, B , of the infimum of $\int_\gamma f$ over minimizing segments γ between them. Since $\int f$ can be made arbitrarily small, some endpoints have this infimum less than one. Choose a minimizing segment for which $\int_\gamma f < 1$; the infimum need not be attained.

A segment meeting $B(q_i, r_i)$, with its endpoints outside $B(q_i, 2r_i)$, spends length at least r_i in the doubled ball; it would have $\int f \geq 1$. The chosen segment therefore misses S . Short paths inside A, B join it to a, b . Letting their radii tend to zero proves that the path-distance infimum in G is at most $d_Z(a, b)$; the reverse inequality holds for every path. The derivative bound of Lemma 3.1 now makes F Lipschitz for d_Z . The regular set is dense and H is complete, so F extends to Z . $\square$

We use two elementary consequences of the small dimension of a removed set. First, in a Euclidean ball or flat cylinder, if $\dim_{\mathbb{H}} E < N - 1$, any two points outside E can be joined outside E by polygonal paths of length arbitrarily close to their distance. The set E need not be closed: intermediate vertices can be chosen arbitrarily near a shortest segment, outside the cones of rays through E from the endpoints. Each bad cone is a countable union of Lipschitz images of E times an interval, and has dimension at most $\dim_{\mathbb{H}} E + 1 < N$. One works in Euclidean lifts for the cylinder.

Second, if E is closed and $\dim_{\mathbb{H}} E < N - 2$, the complement of E in a convex Euclidean ball is simply connected. Indeed a loop has positive distance from E and can be approximated, through a homotopy in the complement, by a polygonal loop. Cone its finitely many edges to a generic point of the ball. For each edge, the bad cone centers lie in countably many bounded Lipschitz images of E times the edge times a real interval: writing an intersection point as $(1 - t)x + tc$ solves for the center c , and one restricts to t bounded away from zero. This bad set has dimension at most $\dim_{\mathbb{H}} E + 2 < N$. A center outside all the bad sets gives the required filling.

Lemma 3.3. *The extension $F : Z \rightarrow H$ is a bi-Lipschitz homeomorphism.*

Proof. Set $E = F(S)$. Its Hausdorff dimension is at most $N - 4$; we will prove closedness only after constructing the inverse on all of H . Local openness of F on G supplies a point of $F(G) \setminus E$. Start there and lift any finite polygonal path in $H \setminus E$ by the local inverse on G . The lift cannot end at a finite parameter: the inverse derivative bound makes it Cauchy, and completeness supplies a limit $z \in Z$. Its image is the prescribed point of the path. This point is outside $F(S)$, hence $z \in G$, where the local inverse continues the lift. Injectivity on G gives uniqueness. The first avoidance fact shows that $H \setminus E \subset F(G)$ and that its inverse K obeys

$$d_Z(K(a), K(b)) \leq C d_H(a, b) \quad (a, b \in H \setminus E).$$

Both the domain and the image of K are dense. The domain is dense by the dimension bound. To see density of the image in Z , use local openness of F on each regular neighborhood and the density of G . Completing K thus gives a Lipschitz map $K : H \rightarrow Z$. The compositions FK and KF equal the identity on dense subsets and hence everywhere. This proves the assertion. It also makes E closed, since S is closed. $\square$

3.3. Removing the apparent singularities. The completion is now the whole cylinder. We next show that its metric is smooth across $E = F(S)$, which will prove that S was empty and permit smooth convergence over every compact cylinder tube.

Transfer the metric on G to $H \setminus E$. It is a flat Kähler metric g_∞ , uniformly bounded above and below by positive multiples of γ . Consider a small convex complex coordinate ball $D \subset H$. Its complement of E is simply connected by the second avoidance fact. A unitary parallel holomorphic coframe therefore exists there:

$$\theta_a = \sum_{j=1}^d A_{aj}(w) dw_j, \quad g_\infty = \sum_a |\theta_a|^2.$$

The coefficient matrix A and its inverse are uniformly bounded, by the two-sided metric bounds.

Each bounded holomorphic coefficient extends holomorphically across E . Here is a direct removal argument. Cover the compact exceptional set in a smaller ball by balls with $\sum r_i^{N-1} \rightarrow 0$, take finite subcovers, and use Lipschitz cutoffs χ_k vanishing near E . Then $\|d\chi_k\|_{L^1} \rightarrow 0$. Testing the Cauchy–Riemann equation for the bounded coefficient against χ_k times a smooth test form and passing to the limit proves the distributional Cauchy–Riemann equation on the whole smaller ball. The distributional equation implies holomorphicity. The inverse bound and continuity show that the extended matrix is nonsingular. The identities expressing that the coframe is closed, and that the resulting metric is flat and Kähler, extend from the dense complement. Thus g_∞ extends as a smooth flat Kähler metric across E . Extensions on overlapping balls agree by density.

The distance of this extended metric is exactly the completion distance already obtained. In fact any piecewise smooth path in a coordinate ball can be approximated in length by paths avoiding E , by polygonal approximation, the avoidance argument, and uniform continuity of the smooth metric. Subdivision gives the assertion for arbitrary paths. Conversely every path off E has its original length. Lemma 3.2 and completion now identify the two distances.

It follows that every point of Z has Euclidean tangent cones, so every point is regular in the original Einstein-limit sense. Thus S is empty. The smooth convergence theorem for Einstein metrics applies at every point of the cylinder, including those not initially known to be regular.

The extended metric is translation invariant. To see this, lift it to $\mathbb{C}^d$, where flatness and simple connectivity give a global parallel unitary holomorphic coframe. Its coefficients in the coordinates w_j are bounded entire functions, again by the uniform metric bounds. Liouville’s theorem makes them constant.

The smooth limit must now be expressed in the original logarithmic coordinates. This will let us perturb a torus while retaining its angular parametrization. Write

$$\mathcal{L}_t(q) = (x_j(q) - x_j(p_t) + i \arg z_j(q))_{j=1}^d \in H, \quad Q_r = \{w \in H : |\operatorname{Re} w| < r\}.$$

Here $r > 0$ is fixed, and $\mathcal{L}_t$ is defined on the part of Y_t in the crossing chart. The real shifts do not alter its angular coordinates. Normalize η_t by a positive scalar to obtain a holomorphic volume form Ω_t with the Calabi–Yau normalization for g_t .

Lemma 3.4 (Convergence in the original logarithmic coordinates). *For every fixed $r > 0$, the chart $\mathcal{L}_t$ contains Q_r in its image for all sufficiently small t along the selected subsequence. Its inverse $\Psi_t : Q_r \rightarrow Y_t$ satisfies*

$$\Psi_t^* g_t \longrightarrow g_\infty, \quad \Psi_t^* \omega_t \longrightarrow \omega_\infty, \quad \Psi_t^* \Omega_t \longrightarrow \Omega_\infty \quad \text{in } C_{\text{loc}}^\infty(Q_r),$$

where the limits are constant tensors and $\Omega_\infty \neq 0$. The angular coordinates of Ψ_t are the original $\arg z_j$.

Proof. The inclusion $Q_r \subset \mathcal{L}_t(Y_t \cap \text{chart})$ follows already from $x(p_t)/s \in B_0 \Subset \Sigma$: a bounded real shift leaves all $d+1$ coordinates $z_0, \dots, z_d$ exponentially small. The issue is to identify these coordinate inverses with the smooth convergence maps.

Take smooth local identifications compatible with the pointed Gromov–Hausdorff convergence, now available at every point of Z . The embedding-map and harmonic-coordinate arguments of Section 2 apply on each such neighborhood. They give smooth local subsequential limits of

$\mathcal{L}_t$. On its intersection with the old regular set, these limits equal F . Indeed, the old and new identifications send each point to points whose g_t -distance tends to zero; on a slightly larger smooth neighborhood, the interior harmonic estimates bound the derivatives of the shifted logarithms and their angular maps. Thus the two limits coincide there. The old regular set is dense, so every local limit equals F throughout its neighborhood.

The derivative of F is nonsingular. The inverse function theorem therefore gives smooth convergence of the local inverse maps on smaller image neighborhoods. Those inverses agree with Ψ_t , because the actual logarithmic chart is injective. A finite cover of each compact subset of Q_r now proves the asserted convergence of the metrics and Kähler forms. This argument uses the actual circle-valued functions $\arg z_j$ at every stage; no change of their angular marking occurs.

Finally, in these coordinates Ω_t is a positive normalizing scalar times $f(z) dw_1 \wedge \cdots \wedge dw_d$. The scalar converges to a positive limit by (2.4). On every fixed tube, $z \rightarrow o$ exponentially and $f(z) \rightarrow f(o)$ with all derivatives in the w coordinates. This gives the asserted nonzero constant limit of Ω_t . The metric limit is translation invariant by the preceding argument, so all three limiting tensors are constant. $\square$

3.4. A special Lagrangian torus in the nearby fibers. Fix a tube Q_{2r} and use the charts of Lemma 3.4 to regard its Calabi–Yau data as tensors on this one fixed domain. We suppress the pullback symbols. The polarizing line bundle extends over the total space and is topologically trivial on a sufficiently small contractible coordinate neighborhood of the maximal crossing. Thus ω_t is exact on Q_{2r} . Its constant limit ω_∞ has zero period on every angular coordinate two-torus. Since its coefficients are constant, every angular coefficient vanishes. Thus

$$T_0 = \{\operatorname{Re} w = 0\} \subset Q_{2r}$$

is Lagrangian for ω_∞ . The constant nonzero form Ω_∞ has constant phase on T_0 , making it special Lagrangian.

We perturb this torus using the persistence argument for Calabi–Yau torus tubes in [48, Section 4, Lemmas 4.3–4.5]. Its linearization is McLean’s special Lagrangian deformation operator [33]. Identify normal vector fields along T_0 with one-forms α by contraction with ω_∞ , and use a fixed normal graph chart $\iota_\alpha : T_0 \rightarrow Q_r$. Choose a sufficiently large integer k and $0 < \beta < 1$. The source space is

$$\mathcal{V} = \{\alpha \in C^{k+1, \beta}(T^*T_0) : \alpha \perp \mathcal{H}^1(T_0)\},$$

where orthogonality and harmonic forms use the flat metric on T_0 . All graphs with α in a fixed sufficiently small ball of $\mathcal{V}$ lie in Q_r .

The periods $\int_{T_0} \Omega_t$ converge to the nonzero flat period. Multiply each Ω_t , and its limit, by the phase making this period positive real. These phases converge. For each small graph set

$$\mathcal{P}_t(\alpha) = (\iota_\alpha^* \omega_t, \iota_\alpha^* \operatorname{Im} \Omega_t).$$

The first component is exact because ω_t is exact. The second has zero integral because graph isotopy preserves $\int_{T_0} \Omega_t$, whose imaginary part vanishes after the phase choice. Thus the target is the fixed Banach space $\mathcal{W}$ consisting of exact $C^{k, \beta}$ two-forms and exact $C^{k, \beta}$ top forms on T_0 . These exact-form spaces are closed.

We have $\mathcal{P}_\infty(0) = 0$. Differentiation of pullbacks of closed forms gives

$$(3.5) \quad D\mathcal{P}_\infty(0)\alpha = (d\alpha, c d(*\alpha)), \quad c \neq 0,$$

where the star is the Hodge star of T_0 . The second identity follows by contraction in an orthonormal Lagrangian frame. This operator is an isomorphism $\mathcal{V} \rightarrow \mathcal{W}$. To see this, rewrite its second equation as $\delta\alpha = f$, with f of mean zero. For an exact two-form ξ , the Hodge Green operator $\mathcal{G}$ gives

$$\alpha = \delta\mathcal{G}\xi + d\mathcal{G}f.$$

The Hodge identities give $d\alpha = \xi$ and $\delta\alpha = f$. The solution is orthogonal to harmonic forms, and the only solution of the homogeneous equations in $\mathcal{V}$ is zero. Elliptic estimates make the inverse bounded, with one derivative of gain.

Lemma 3.4 gives $\mathcal{P}_t \rightarrow \mathcal{P}_\infty$ in C^1 on a fixed small ball of $\mathcal{V}$, with values in $\mathcal{W}$. In particular, $\mathcal{P}_t(0) \rightarrow 0$, and their derivatives are uniformly close to the fixed invertible operator (3.5) after the ball is made smaller. The inverse function theorem therefore produces $\alpha_t \rightarrow 0$ in $C^{k+1,\beta}$ with $\mathcal{P}_t(\alpha_t) = 0$ for every sufficiently late member of the subsequence. This argument requires convergence of the ambient tensors, not differentiability with respect to the degenerating parameter. On each connected Lagrangian graph, the real restriction of Ω_t is nowhere zero. Its sign is therefore constant, and the positive period makes it positive. Thus the graph is calibrated by $\text{Re } \Omega_t$ and hence minimal. The minimal-graph equation is elliptic; smoothness of the ambient data and elliptic regularity therefore make each graph smooth.

The graphs are embedded, and $\iota_{u\alpha_t}$, $0 \leq u \leq 1$, is an isotopy to T_0 inside Q_r . The actual chart Ψ_t transports this isotopy to the smooth fiber, ending at an angular coordinate torus in the prescribed crossing chart. The graph is Lagrangian and has constant volume-form phase, so it is the torus required by Theorem 2.1.

The initial sequence was arbitrary. If the conclusion failed for arbitrarily small nonzero parameters, choose $t_\nu \rightarrow 0$ so that no Y_{t_ν} contained a torus with the stated properties. The preceding construction gives such tori on a tail of a subsequence, a contradiction. This proves the assertion for every sufficiently small nonzero parameter and completes the proof of Theorem 2.1.

4. A POLARIZED DEGENERATION WITH THE PRESCRIBED ISOTROPIC CLASS

The degeneration must supply two kinds of data. Its volume growth and a nonzero residue at a maximal crossing will allow us to apply Theorem 2.1. Its marked periods and second-cohomology monodromy will retain the prescribed isotropic class for rotation. We construct both in the same polarized family. All markings take values in

$$\Lambda = H^2(X, \mathbb{Z}), \quad \Lambda_{\mathbb{Q}} = \Lambda \otimes_{\mathbb{Z}} \mathbb{Q}, \quad r = \text{rank } \Lambda,$$

with its integral Beauville–Bogomolov–Fujiki form q . We write $q(x, y)$ for the associated symmetric bilinear form, so that $q(x) = q(x, x)$. Markings are obtained by deformation and parallel transport from the differentiable manifold underlying X . Fix the connected Teichmüller component containing X . Along deformation paths use Ehresmann trivializations anchored at X ; the transported complex structures on its differentiable manifold then stay in this component.

The universal coefficient theorem and $H_1(X, \mathbb{Z}) = 0$ show that Λ is torsion-free. Write $c_1(L) = m_0 e$ with $m_0 \in \mathbb{Z}_{>0}$ and $e \in \Lambda$ primitive. Since $H^1(X, \mathcal{O}_X) = 0$ by Hodge theory, the exponential sequence identifies $\text{Pic}(X)$ with the integral $(1, 1)$ classes. Thus $e = c_1(E)$ for a unique line bundle E , and $L \simeq E^{\otimes m_0}$. The class e is nonzero, nef, and isotropic. It will remain the same labelled lattice vector, although it need not remain of type $(1, 1)$ during the construction. Its sign is fixed by

$$q(e, \kappa_X) > 0$$

for every Kähler class κ_X on X : in the Lorentzian space $H^{1,1}(X, \mathbb{R})$, a nonzero null vector in the closure of the positive cone pairs positively with every vector in that cone.

Proposition 4.1 (The prescribed-cusp family). *Put $d = 2n$. There exist a primitive vector $h \in \Lambda$ with $q(h) > 0$ and $q(h, e) = 0$, a rational subspace*

$$P = \langle e, f, v \rangle_{\mathbb{Q}} \subset h^{\perp}, \quad \text{sign}(q|_P) = (2, 1),$$

and a projective semistable family $\pi : \mathcal{Y} \rightarrow \Delta$ with the following properties. Here Δ is a sufficiently small disk in a smooth algebraic curve, and $\Delta^ = \Delta \setminus \{0\}$.*

(1) *The total space $\mathcal{Y}$ is smooth, the central fiber is a reduced simple normal crossings divisor, and every Y_t , $t \neq 0$, is an irreducible holomorphic symplectic manifold in the marked deformation component of X . A line bundle $\mathcal{A}$ on $\mathcal{Y}$ restricts to an ample bundle on every Y_t with $c_1(\mathcal{A}|_{Y_t}) = kh$ for a fixed positive integer k under parallel transport. The family is the analytic restriction of an algebraic family.*

(2) *The rational vectors can be chosen so that*

$$(4.1) \quad q(e) = q(f) = 0, \quad q(e, f) = 1, \quad q(e, v) = q(f, v) = 0, \quad a := q(v) > 0.$$

On the universal cover of Δ^ the marked Hodge line is spanned by the normalized vector*

$$(4.2) \quad p(\tau) = f + \tau v - \frac{a}{2}\tau^2 e, \quad \text{Im } \tau > 0.$$

The function τ tends to the cusp represented by the original line $\mathbb{Q}e$, and

$$(4.3) \quad q(p(\tau)) = 0, \quad q(p(\tau), e) = 1, \quad q(p(\tau), \overline{p(\tau)}) = 2a(\text{Im } \tau)^2.$$

For a consistent convention for second-cohomology monodromy there is a rational $w > 0$ such that

$$(4.4) \quad Me = e, \quad Mv = v - awe, \quad Mf = f + wv - \frac{aw^2}{2}e, \quad M|_{P^{\perp}} = 1.$$

The orthogonal complement in the last formula is taken in $\Lambda_{\mathbb{Q}}$. In particular,

$$(4.5) \quad (M - 1)P = \langle e, v \rangle_{\mathbb{Q}}, \quad (M - 1)^2 f = -aw^2 e \neq 0, \quad (M - 1)^3 = 0.$$

The same image statement holds if the monodromy convention uses M^{-1} .

(3) *There is a single-valued holomorphic relative symplectic form σ_t on the smooth family, uniquely normalized by $q([\sigma_t], e) = 1$. The forms $\eta_t = \sigma_t^n$ extend to a holomorphic section of the relative logarithmic top-form line bundle of the semistable family. With $s = -\log |t|$,*

$$(4.6) \quad cs^d \leq \int_{Y_t} |\eta_t|^2 \leq Cs^d$$

for positive constants c, C . There is a point $o \in Y_0$ where $d + 1$ components meet and coordinates satisfying

$$(4.7) \quad t = z_0 \cdots z_d, \quad \eta = F(z) \frac{dz_1}{z_1} \wedge \cdots \wedge \frac{dz_d}{z_d}, \quad F(o) \neq 0.$$

(4) *For any prescribed irrational number λ , there is a path tending to 0 in Δ^* , with a continuous choice of lift, along which $\tau = \lambda + iy$ and $y \rightarrow +\infty$.*

The prescribed line $\mathbb{Q}e$ and the normalization $q(p, e) = 1$ are the data needed for rotation. We obtain them in three stages. First, choose a positive polarization perpendicular to e and realize an open set of its periods by an algebraic family. Next, find a rational three-dimensional period subspace containing e and follow its cusp inside that family, retaining the original marking. Finally, compute the periods at the cusp and use their growth to produce the nonzero logarithmic residue required by Theorem 2.1.

4.1. A positive polarization orthogonal to the null class. We use the following standard inputs from hyperkähler period theory: unobstructed deformations and local Torelli; surjectivity of the period map from the selected connected marked component; the projectivity criterion for a positive integral $(1, 1)$ class; and the Kähler-cone description, whose walls are orthogonal to negative integral $(1, 1)$ classes. See [21, 22, 44, 45]. The corrected projectivity criterion is [22, Theorem 2], and the Kähler-cone description in terms of negative MBM classes is due to Amerik–Verbitsky [1, Theorems 1.19 and 6.2]; see also [40, Section 2.1].

The first two construction steps also have an independent use: they provide an algebraic family and a quotient in which equality of period lifts detects equality of monodromy on the full second-cohomology lattice. We isolate precisely this construction, before choosing a ternary subdomain or a cusp.

Lemma 4.2 (A positive polarization and a free period quotient). *Let X be a compact irreducible holomorphic symplectic Kähler manifold, fix its connected marked deformation component, and put $\Lambda = H^2(X, \mathbb{Z})$ with its integral BBF form q . For each nonzero primitive $e \in \Lambda$ with $q(e) = 0$, there are*

- (1) *a primitive $h \in \Lambda$ with $q(h) > 0$ and $q(h, e) = 0$, and a manifold in the chosen marked component on which h is ample;*
- (2) *a smooth projective algebraic family over a smooth connected quasi-projective base S , whose fibers are irreducible holomorphic symplectic manifolds in that component, with a relative polarization of marked class kh for one $k \in \mathbb{Z}_{>0}$;*
- (3) *a torsion-free finite-index subgroup*

$$\Gamma \subset G_h(\mathbb{Z}) := \{g \in O(\Lambda, q) : gh = h, gD_h = D_h\},$$

where D_h is the component of the polarized period domain containing the chosen ample period, such that the family monodromy is contained in Γ and the descended period map

$$\Phi : S \longrightarrow Q_h := \Gamma \backslash D_h$$

is algebraic. The quotient Q_h is smooth and quasi-projective, $D_h \rightarrow Q_h$ is a covering map with free Γ -action, and a local marked lift of Φ is submersive at some point of S .

More generally, conclusions (2) and (3) hold for any prescribed primitive positive $h \in \Lambda$ which is ample on a manifold in the chosen marked component. The markings are obtained by deformation and parallel transport from X . No finite-index assertion about the image of the family monodromy inside Γ is part of this construction.

Proof. The positive polarization. The signature of q is $(3, r - 3)$. The existence of the nonzero null vector e implies $r \geq 4$. The radical of $e^\perp$ is $\mathbb{R}e$, and

$$\text{sign}(e^\perp / \mathbb{R}e) = (2, r - 4).$$

Thus $e^\perp$ contains positive real vectors. Since e is rational, $e^\perp$ is defined over $\mathbb{Q}$; rational approximation inside this space supplies a rational positive vector. Taking the primitive integral generator of its ray gives h with $q(h) > 0$ and $q(h, e) = 0$.

Set $V = h^\perp \subset \Lambda_{\mathbb{Q}}$; its signature is $(2, r - 3)$. The polarized period domain is

$$\{[p] \in \mathbb{P}(V_{\mathbb{C}}) : q(p) = 0, q(p, \bar{p}) > 0\}.$$

A generic point of this domain is orthogonal to no integral class outside $\mathbb{Q}h$: the excluded sets are countably many proper hyperplane sections. Period surjectivity realizes such a point in the selected marked component. Its Néron–Severi lattice is $\mathbb{Z}h$. One of $h, -h$ is in the component of the positive cone containing the Kähler cone. There are no negative integral $(1, 1)$ classes on this manifold, so the Kähler-cone description makes that choice a Kähler class. After replacing h by $-h$ if necessary, it is the class of an ample bundle. This sign change does not change V or e . Fix

the resulting polarized manifold and let D_h denote the component of the polarized period domain containing its period. Locally, the periods of its polarized deformations fill an open subset of D_h . *The algebraic family and the free arithmetic quotient.*

We need an algebraic family with an open marked period image. Choose $k \in \mathbb{Z}_{>0}$ so that the bundle with class kh on the preceding manifold is very ample and has vanishing higher cohomology. The bundle extends locally over polarized deformations; bases of its sections give nearby embeddings with the same Hilbert polynomial. Locally, deformations of a complete-linear-system embedding are deformations of the polarized pair together with projective frames of these section spaces. The pair deformation space is smooth by local Torelli, and cohomology vanishing lets the frames vary freely. There are no infinitesimal automorphisms, and finite automorphisms act freely on frames. A reduced irreducible Hilbert-scheme component therefore has a smooth point with submersive polarized period map.

Restrict its universal family to a smooth connected Zariski-open base S of smooth irreducible holomorphic symplectic fibers, retaining a submersive point. This restriction is algebraic: Hodge numbers are locally constant in a smooth projective family, and the degeneracy locus of the relative two-form has closed image in the base by properness. Simple connectivity and differentiable type are preserved in the connected smooth family. The universal hyperplane bundle has class kh .

Let

$$G_h(\mathbb{Z}) = \{g \in O(\Lambda, q) : gh = h, gD_h = D_h\}.$$

Choose a torsion-free finite-index congruence subgroup $\Gamma \subset G_h(\mathbb{Z})$ whose elements have determinant one. Both groups act on the full lattice Λ . Restriction embeds $G_h(\mathbb{Z})$ in $O(V, q)$, because $\Lambda_{\mathbb{Q}} = \mathbb{Q}h \oplus V$; its image is arithmetic. For example, a congruence subgroup acting trivially on the finite gluing quotient between $\mathbb{Z}h \oplus (\Lambda \cap V)$ and Λ extends across this quotient, which shows the required finite-index property in the integral orthogonal group of V .

The family monodromy is contained in $G_h(\mathbb{Z})$: it preserves the BBF form and the class kh , hence also h , and preserves the component of its periods. Pass to the finite cover of S corresponding to the inverse image of Γ under this monodromy representation. A finite topological cover of a smooth complex quasi-projective variety is algebraic and finite étale [16, Exposé XII, Theorem 5.1]. We keep one connected component and continue to call the base S . This step requires only that Γ has finite index in $G_h(\mathbb{Z})$; no hypothesis on the index of the family monodromy image is used.

The action of Γ on D_h is free. Indeed, the stabilizer of a positive oriented two-plane is a compact subgroup of $O(V_{\mathbb{R}}, q)$. Its intersection with a discrete integral subgroup is finite and is therefore trivial in the torsion-free group Γ . This argument also removes any kernel of the projective action. Consequently

$$Q_h := \Gamma \backslash D_h$$

is a smooth arithmetic quotient and $D_h \rightarrow Q_h$ is a covering map. The Baily–Borel theorem [2] makes Q_h quasi-projective. Borel’s algebraicity theorem [5, Theorem 3.10] then makes the descended period map

$$\Phi : S \longrightarrow Q_h$$

algebraic. We have given the Hilbert-family construction directly; compare [39, Lemma 4.5]. At a chosen point $s_0 \in S$, fix the marking obtained by transport from the starting polarized manifold and a local lift of Φ to D_h . After shrinking an analytic neighborhood, this lift is submersive and its image contains an open set $U \subset D_h$ with the actual marking. Freeness will later identify monodromy on the full second-cohomology lattice from equality of these lifted periods. $\square$

We now apply Lemma 4.2 to the primitive class fixed at the start of this section. Its open marked period image is the input for the next step.

4.2. A rational ternary domain containing the fixed vector. We now find a rational period subspace which meets U and has $\mathbb{Q}e$ as a cusp. View D_h as the space of positive oriented real two-planes in $V_{\mathbb{R}}$. Let $W \in U$. Write $e = e_W + e_{\perp}$ in the orthogonal decomposition $V_{\mathbb{R}} = W \oplus W^{\perp}$. Since $W^{\perp}$ is negative definite, $e_W \neq 0$; otherwise the nonzero vector e would have negative square. Moreover

$$q(e_{\perp}) = -q(e_W) < 0.$$

It follows that $W + \mathbb{R}e = W \oplus \mathbb{R}e_{\perp}$ has signature $(2, 1)$.

For an explicit rational approximation, choose an oriented basis x, y of W and rational vectors $x_j, y_j \in V$ tending to x, y . For all sufficiently large j the plane $W_j = \langle x_j, y_j \rangle_{\mathbb{R}}$ is positive, has the chosen orientation, and lies in U . Nondegeneracy and signature are open conditions on the Gram matrix, so

$$P_j = \langle e, x_j, y_j \rangle_{\mathbb{Q}}$$

has signature $(2, 1)$ for all sufficiently large j . Fix such a $P = P_j$. The vector e has remained fixed throughout this approximation. Let D_P be the component of its positive-plane domain that contains W_j . Then $D_P \subset D_h$ and $D_P \cap U \neq \emptyset$.

An arithmetic subgroup of $SO(P, q)$ preserving D_P can be made to act on Λ by extending it identically on $P^{\perp}$. To see the integral issue explicitly, set

$$\Lambda_0 = (P \cap \Lambda) \oplus (P^{\perp} \cap \Lambda).$$

This is a finite-index sublattice of Λ . Choose N with $N\Lambda \subset \Lambda_0$. If an isometry g of $P \cap \Lambda$ is congruent to the identity modulo mN , with $m \in \mathbb{Z}_{>0}$, its extension satisfies $(g - 1)\Lambda \subset m\Lambda_0 \subset m\Lambda$. The same holds for its inverse, so the extension preserves Λ ; choosing m sufficiently divisible places it in Γ . Take such a finite-index subgroup Γ_P . It is torsion-free and its inclusion into Γ is an inclusion of actual lattice isometries.

The quotient

$$C_P = \Gamma_P \backslash D_P$$

is a smooth arithmetic curve. Its holomorphic map $j : C_P \rightarrow Q_h$ is algebraic by Borel's theorem [5, Theorem 3.10]. Since P contains the rational null line $\mathbb{Q}e$, its smooth compactification $\overline{C}_P$ has a cusp c_e represented by that line. We always retain this chosen lift of the cusp in the boundary of D_P , even if other translates represent the same point of $\overline{C}_P$.

The construction also covers the two smallest possible ranks. If $r = 4$, then V itself is a rational three-space of signature $(2, 1)$ and we simply take $P = V$. Its noncompact arithmetic curve has a cusp because the given e is rational and isotropic. If $r = 5$, the domain of signature $(2, 2)$ is still a Hermitian symmetric domain and the same ternary subdomain works. General constructions that obtain a null vector from Meyer's theorem need an additional rank hypothesis; here e is already supplied.

4.3. An algebraic curve reaching the specified cusp. It remains to reach this cusp in the actual polarized family and identify its monodromy. Form the algebraic fiber product

$$Z = S \times_{Q_h} C_P.$$

Choose a point z_0 at which the marked period lift from S equals the chosen point of $U \cap D_P$. The projection $Z \rightarrow C_P$ is submersive near z_0 , because the local period map from S is submersive. Thus the component through z_0 dominates C_P . It contains an irreducible algebraic curve through a point in this selected local branch which dominates C_P : intersect a quasi-projective embedding with general hyperplanes through a smooth point, choosing their tangent intersection transverse to the fiber of $Z \rightarrow C_P$, and take the component through that point. Normalize this curve and call it B . At the selected point the resulting map $g : B \rightarrow C_P$ is nonconstant. Pullback from S gives a projective polarized family over B with the chosen marked local branch.

Take smooth projective completions. The nonconstant map of curves extends to a finite surjective morphism

$$\bar{g}: \bar{B} \longrightarrow \bar{C}_P.$$

There is a finite set $F \subset \bar{C}_P$, containing all cusps, branch values, and images of the finitely many points omitted from B , such that over $C_P \setminus F$ the map is a finite unramified covering and its source lies in B . If necessary move the chosen starting point a little within the same local branch so that it maps outside F .

The following path choice fixes the marking at infinity. Start with its selected lift in D_P . The inverse image in D_P of $F \cap C_P$ is a locally finite discrete set. Its complement is path connected. Consequently we may take a path from the initial lift towards the boundary point represented by the specific vector e , avoiding that discrete set; sufficiently far along the path it lies in a cusp neighborhood of e . Project the path to $C_P \setminus F$ and lift it through the covering from B , starting at the chosen point. The lift approaches some point $b_e \in \bar{B}$ over c_e . In a punctured disk around b_e we therefore have the desired smooth algebraic polarized family.

Transport the original marking along this lifted path. The resulting family period lift to D_h and the inclusion of the chosen D_P -valued lift start at the same point and project to the same map into Q_h . Uniqueness of lifts for the covering $D_h \rightarrow Q_h$ makes them equal all along the path and on the universal cover of the resulting punctured disk.

Freeness now identifies the monodromy on all of Λ . For a loop in B , let $M_B \in \Gamma$ be its second-cohomology monodromy and let $\gamma_B \in \Gamma_P \subset \Gamma$ be the deck transformation of its modular-curve lift, using compatible conventions. Equivariance and the preceding equality of period lifts imply

$$M_B \tilde{\Phi}(\tilde{b}) = \gamma_B \tilde{\Phi}(\tilde{b}).$$

Hence $\gamma_B^{-1} M_B$ fixes a point of D_h . The action is free, so $M_B = \gamma_B$ as isometries of the full lattice. In particular, near the selected cusp the actual family monodromy fixes our labelled e and is the identity on $P^\perp$. Thus the free action and the equality of the initial marked lifts retain the precise class needed in the rotation argument.

4.4. The exact period and monodromy formulas. We have obtained a family whose marked periods lie in D_P and approach the cusp $\mathbb{Q}e$. A rational hyperbolic basis makes both their growth and their monodromy explicit. Choose $f_0 \in P$ with $q(e, f_0) = 1$ and put $f = f_0 - \frac{1}{2}q(f_0)e$. Then $q(f) = 0$ and $q(e, f) = 1$. The orthogonal complement of $\langle e, f \rangle$ in P is a positive rational line; choose a nonzero rational vector v on it and put $a = q(v) > 0$. This proves (4.1).

For every positive period in $P_\mathbb{C}$ the pairing with e is nonzero. Indeed, a positive real two-plane orthogonal to e would put e in its negative-definite orthogonal complement. Normalize the period by $q(p, e) = 1$. Writing $p = f + \tau v + \beta e$, its nullity gives $\beta = -a\tau^2/2$. Positivity becomes

$$q(p, \bar{p}) = a|\tau|^2 - \frac{a}{2}(\tau^2 + \bar{\tau}^2) = 2a(\operatorname{Im} \tau)^2 > 0.$$

Change the sign of v so that the component D_P is parametrized by $\operatorname{Im} \tau > 0$. The point $\tau = \infty$ represents the line $\mathbb{Q}e$.

For a rational number u , define an isometry U_u of P by

$$U_u e = e, \quad U_u v = v - aue, \quad U_u f = f + uv - \frac{au^2}{2}e.$$

The Gram matrix in (4.1) verifies directly that U_u preserves q , and substitution gives

$$U_u p(\tau) = p(\tau + u).$$

Taking u to be a sufficiently divisible positive integer makes this transformation belong to Γ_P . The cusp stabilizer in the torsion-free orientation-preserving arithmetic group is generated by such

a translation $\tau \mapsto \tau + w_0$, with $w_0 > 0$. Here the width w_0 is rational because the basis e, f, v is rational and the generator preserves that rational structure. The standard cusp coordinate is

$$u_c = \exp(2\pi i \tau / w_0).$$

If a local parameter t upstairs has ramification degree b at the cusp, then

$$(4.8) \quad u_c = t^b \beta(t), \quad \beta(0) \neq 0, \quad \text{Im } \tau = \frac{bw_0}{2\pi} (-\log |t|) + O(1).$$

The error is uniform near the puncture. A positive local loop corresponds to U_{bw_0} . The already established equality of monodromy representations gives (4.4) with $w = bw_0$. Computing $(U_w - 1)$ on the displayed basis gives (4.5). Further finite base changes replace w by a positive integral multiple, so all the image and nonvanishing statements survive.

Because only finitely many exceptional values were removed from $\overline{C}_P$, a sufficiently small punctured cusp neighborhood contains none of them. In the preceding path construction we may therefore choose its final lift in D_P to be the ray $\tau = \lambda + iy$ for any specified irrational λ and all large y . The ray lifts to the chosen finite cover after its initial branch is fixed. It continues to lift after any further finite base change. The resulting path in the final parameter t may be nonradial.

4.5. Semistable reduction and the logarithmic form. Take a projective algebraic model over a neighborhood of b_e in $\overline{B}$. One-parameter semistable reduction in characteristic zero, followed by restriction to a small disk, gives, after finite base change and a projective modification which is an isomorphism over the punctured base, a smooth total space $\mathcal{Y}$ with reduced simple normal crossings central fiber; see [23, Chapter II, Semi-stable Reduction Theorem, pp. 53–54]. The smooth fibers and their marked periods remain those already constructed. The polarization on the punctured family extends to a line bundle on $\mathcal{Y}$: choose a rational section, close its divisor in the smooth total space, and use that every Weil divisor on a smooth variety is Cartier. Denote the extension by $\mathcal{A}$. Its restriction to each smooth fiber is ample; relative ampleness on the semistable model is unnecessary.

The invariant vector e is now a single-valued flat section of $R^2\pi_*\mathbb{Z}$ on Δ^* . The pairing $q(-, e)$ is a single-valued holomorphic functional on its associated flat vector bundle. Its restriction to the Hodge line is nowhere zero, by (4.3). Therefore it trivializes that Hodge line. Holomorphic base change identifies the Hodge line with the bundle of relative holomorphic two-forms, so there is a unique holomorphic relative form σ with $q([\sigma], e) = 1$. On the universal cover its class is exactly $p(\tau)$, including its scalar. Thus $\eta = \sigma^n$ is a single-valued holomorphic relative volume form on Δ^* .

Polarizing the Fujiki identity [13], or simply comparing the coefficient of $z^n w^n$ in its application to $z[\sigma] + w[\bar{\sigma}]$, gives

$$\binom{2n}{n} \int_{Y_t} \sigma_t^n \wedge \bar{\sigma}_t^n = c_X 2^n q(p(\tau), \overline{p(\tau)})^n.$$

For the positive volume convention in $|\eta_t|^2$, any fixed normalization factor can be absorbed in a constant. Equations (4.3) and (4.8) consequently give

$$\int_{Y_t} |\eta_t|^2 = c' (\text{Im } \tau)^{2n} \asymp s^{2n} = s^d, \quad c' > 0,$$

proving (4.6) uniformly near the puncture.

Let $\mathcal{K}_{\log}$ denote the invertible sheaf

$$\det \left(\Omega_{\mathcal{Y}}^1(\log Y_0) / \pi^* \Omega_{\Delta}^1(\log \{0\}) \right).$$

The quotient is locally free of rank d in the semistable coordinates. On Δ^* its determinant is the ordinary relative canonical line, and η is a section of it there. We prove that η extends holomorphically as a section of $\mathcal{K}_{\log}$.

First consider a point of Y_0 lying on only one component. There the family has product coordinates $(z_1, \dots, z_d, t)$. Write

$$\eta_t = F(z, t) dz_1 \wedge \dots \wedge dz_d.$$

On a smaller product chart the submean inequality in the fiber variables, followed by the global upper volume bound, gives

$$|F(z, t)|^2 \leq C \int_{Y_t} |\eta_t|^2 \leq C' s^d.$$

For $m \geq 1$ the negative Laurent coefficient in the t variable satisfies, uniformly on a smaller z -polydisk,

$$\left| \frac{1}{2\pi i} \int_{|t|=\rho} F(z, t) t^{m-1} dt \right| \leq C \rho^m |\log \rho|^{d/2} \rightarrow 0.$$

All negative Laurent coefficients vanish. Hence F extends holomorphically through $t = 0$. This extends η over the smooth locus of the reduced central divisor. The crossing locus has complex codimension at least two in the smooth total space. Local trivializations of $\mathcal{K}_{\log}$ and Hartogs' extension theorem then extend η uniquely over that locus. Thus the polynomial bound in s suffices for logarithmic extension.

4.6. Maximal growth forces a nonzero maximal residue. We have the semistable family and the extended logarithmic form. To apply Theorem 2.1, we must still find a maximal crossing with nonzero coefficient. The lower growth bound forces one, as follows. At a point where precisely k components of Y_0 meet, take coordinates with

$$t = z_0 \cdots z_{k-1}, \quad 1 \leq k \leq d+1.$$

A local generator of $\mathcal{K}_{\log}$ restricts to the fiber as

$$\frac{dz_1}{z_1} \wedge \dots \wedge \frac{dz_{k-1}}{z_{k-1}} \wedge dz_k \wedge \dots \wedge dz_d.$$

Its extended holomorphic coefficient is bounded on a smaller chart. In the variables $x_i = -\log |z_i|$, $1 \leq i \leq k-1$, the fiber part of a fixed coordinate polydisk has an expanding simplex as its radial domain: each x_i is bounded below, and $\sum_i x_i \leq s + O(1)$. Its dimension is $k-1$. Integrating the angular variables and the remaining ordinary coordinates therefore yields

$$(4.9) \quad \int_{Y_t \cap \text{chart}} |\eta_t|^2 \leq C(1 + s^{k-1}).$$

For $k = d+1$, the crossing is an isolated point o . Suppose its coefficient F vanishes at o . To improve (4.9), choose a polydisk $|z_i| < \delta < 1$, put $A = -\log \delta$, and write

$$x_i = A + (s - (d+1)A)u_i, \quad u_i > 0, \quad \sum_{i=1}^d u_i < 1.$$

For almost every u in this fixed simplex, all $d+1$ coordinates $z_0, \dots, z_d$ tend to zero as $s \rightarrow \infty$, uniformly in their arguments. The rescaled integrand $|F|^2$ thus tends to $|F(o)|^2 = 0$ almost everywhere. It is uniformly bounded and the simplex and angle torus have finite measure. Dominated convergence gives

$$(4.10) \quad \int_{Y_t \cap \text{chart}} |\eta_t|^2 = o(s^d) \quad \text{if } F(o) = 0.$$

Properness gives finitely many such charts covering the central fiber and, after shrinking the base, all nearby fibers. If there were no maximal crossings, every chart would contribute $O(s^{d-1})$. If maximal crossings existed but all their coefficients vanished, their contributions would be $o(s^d)$ by (4.10), with all other contributions again $O(s^{d-1})$. Either conclusion contradicts the lower

bound in (4.6). At least one maximal crossing therefore has nonzero coefficient. This proves (4.7). At that point the functions $z_1, \dots, z_d$ may be chosen to be algebraic local defining functions of the corresponding components; the unit in the product equation is absorbed in z_0 . They are thus restrictions of rational functions on the algebraic total space, as required for the intersection estimates in Theorem 2.1. This completes the proof of Proposition 4.1.

Corollary 4.3. *Fix an irrational λ and choose the lifted path in Proposition 4.1 with $\tau = \lambda + iy$. For every sufficiently large y , the corresponding smooth fiber contains an embedded special Lagrangian real d -torus for its Ricci-flat metric in the polarization class. This torus is isotopic to a coordinate angle torus in the chart (4.7). The period and monodromy on that fiber are the exact marked data (4.2)–(4.5).*

Proof. The algebraicity, projectivity, extension of the polarization, volume growth, logarithmic extension, and nonzero coefficient required by Theorem 2.1 are all supplied by Proposition 4.1. The theorem applies to every sufficiently small parameter, in particular along the chosen path, where $\operatorname{Re} \tau = \lambda$ and the marking is already fixed. $\square$

5. ROTATION AND RETURN TO THE ORIGINAL MANIFOLD

We now complete the proof of Theorem 1.1. The marking in Proposition 4.1 will be used throughout: all second cohomology groups are identified with the lattice of the differentiable manifold underlying the given X , and q denotes both the quadratic form and its associated symmetric bilinear form.

5.1. The primitive class and the chosen fiber. Recall the primitive class e and line bundle E constructed at the start of Section 4. Write $E_X = E$, so that $L \simeq E_X^{\otimes m_0}$. The class e is nonzero, nef, and $q(e) = 0$; its sign was fixed so that, for every Kähler class κ on X ,

$$(5.1) \quad q(e, \kappa) > 0.$$

Fix $\lambda \in \mathbb{R} \setminus \mathbb{Q}$ and use the family of Proposition 4.1. Its rational classes e, f, v, h satisfy

$$q(e) = q(f) = 0, \quad q(e, f) = 1, \quad q(e, v) = q(f, v) = 0, \quad a := q(v) > 0,$$

and $h \perp P := \langle e, f, v \rangle_{\mathbb{Q}}$. The normalized period and the local monodromy M on H^2 are

$$(5.2) \quad p(\tau) = f + \tau v - \frac{a}{2} \tau^2 e, \quad \tau = \lambda + iy, \quad y > 0,$$

$$(5.3) \quad Me = e, \quad Mv = v - awe, \quad Mf = f + wv - \frac{aw^2}{2}e, \quad w \neq 0.$$

Consequently,

$$(5.4) \quad (M - 1)P = \langle e, v \rangle_{\mathbb{Q}}.$$

Theorem 2.1, applied along the prescribed vertical ray, provides a smooth fiber $Y = Y_t$, its polarized Ricci-flat metric g , and an embedded special Lagrangian torus $T \subset Y$. Here $\dim_{\mathbb{C}} Y = 2n$, $\dim_{\mathbb{R}} T = 2n$, and T is isotopic to a coordinate torus in a semistable chart $z_0 z_1 \cdots z_{2n} = t$. A constant rescaling of g , as used in the torus construction, does not affect these properties.

Write the present complex structure as I . Choose a hyperkähler triple (I, J, K) for g , with $IJ = K$, so that

$$(5.5) \quad \omega_J + i\omega_K = c\sigma_t, \quad c > 0, \quad [\omega_I] \in \mathbb{R}_{>0}h.$$

The positive scalar normalizes the length of σ_t ; its given phase fixes the ordered pair (J, K) . Since $[\sigma_t] = p(\tau)$,

$$(5.6) \quad [\omega_K] = cy(v - a\lambda e), \quad q(e, [\omega_J]) = c > 0.$$

The second equality follows from $q(e, p(\tau)) = 1$ and will retain the required sign of e after rotation.

5.2. Restriction, calibration, and complex rotation. The following criterion isolates the rotation argument. Cohomological vanishing is enough because the special Lagrangian condition turns volume into a period.

Lemma 5.1 (A cohomological rotation criterion). *Let (Y, g, I, J, K) be a hyperkähler manifold of real dimension $4n$, where $n \geq 1$ and $IJ = K$, and let $i : T \hookrightarrow Y$ be a compact, connected, oriented embedded special Lagrangian submanifold for the complex structure I . If $i^*[\omega_K] = 0$ in $H^2(T, \mathbb{R})$, then T is a complex n -dimensional submanifold for J and is Lagrangian for $\omega_K + i\omega_I$. If T is diffeomorphic to a real $2n$ -torus, then it is a complex torus for J .*

Proof. The normalized parallel holomorphic volume form for (Y, I, g) is

$$\Omega_I = \frac{(\omega_J + i\omega_K)^n}{n!}.$$

To check the normalization, in a symplectic unitary frame one has $\omega_J + i\omega_K = \sum_{j=1}^n dz_{2j-1} \wedge dz_{2j}$; its n th power divided by $n!$ is the unit holomorphic volume form. It has absolute value one on an orthonormal frame of any I -Lagrangian real $2n$ -plane. The special Lagrangian calibration identity [17, III.1, Theorem 1.10 and Corollary 1.11] therefore gives

$$\text{vol}_g(T) = \left| \int_T \Omega_I \right|.$$

All terms containing ω_K in this integral vanish: the hypothesis makes $i^*\omega_K$ exact, while all the factors in the expansion are closed. Stokes' theorem then yields

$$(5.7) \quad \text{vol}_g(T) = \left| \int_T \frac{\omega_J^n}{n!} \right| \leq \int_T \left| \frac{i^*\omega_J^n}{n!} \right| \leq \text{vol}_g(T).$$

The last inequality is the pointwise Wirtinger inequality [17, II.6, Theorem 6.11]. Its continuous, nonnegative deficit has integral zero, so equality holds at every point of T . Equality planes are exactly the J -invariant real $2n$ -planes. Thus T is a complex n -dimensional submanifold of

$$Y' = (Y, J).$$

Its orientation may now be chosen to be the complex orientation; the absolute period in (5.7) made the initial choice immaterial.

For J the holomorphic symplectic form is $\sigma_J = \omega_K + i\omega_I$. We already have $\omega_I|_T = 0$ pointwise. Since $JT_x T = T_x T$, for $u, v \in T_x T$ the quaternion identities give

$$\omega_K(u, v) = g(IJu, v) = \omega_I(Ju, v) = 0.$$

Hence $\sigma_J|_T = 0$: the rotated submanifold is holomorphically Lagrangian.

Suppose now that T is diffeomorphic to a real $2n$ -torus. The restriction of ω_J makes T Kähler. Consequently $b_1(T) = 2n$ and $h^{1,0}(T) = n$. Its Albanese map

$$\alpha : T \longrightarrow \text{Alb}(T)$$

has a complex n -torus as target and induces an isomorphism on integral first cohomology. Both integral cohomology rings are exterior algebras on first cohomology, so α^* is an isomorphism in every degree. In particular, α has nonzero degree; holomorphicity makes that degree positive, hence equal to one, and α is surjective.

Choose a Kähler class ξ on T . Surjectivity on $H^2(-, \mathbb{R})$ writes $\xi = \alpha^*\beta$ for some real class β on $\text{Alb}(T)$. If a fiber contained a positive-dimensional irreducible compact analytic subvariety V of dimension k , then

$$\int_V \xi^k = 0,$$

since $\alpha|_V$ is constant. This contradicts Kähler positivity. Thus α is proper with finite fibers, hence finite. A finite degree-one holomorphic map to a normal complex space is an isomorphism. This proves that T is a holomorphic Lagrangian complex torus in Y' . $\square$

We verify the criterion for the torus constructed above, using its coordinate isotopy class. This is where the monodromy information is needed.

For a coordinate torus $T_0 = (\mathbb{R}/2\pi\mathbb{Z})^{2n}$ choose its fixed radii $r_1, \dots, r_{2n}$. Around the circle $t(\theta) = t \exp(i\theta)$, the formula

$$z_j = r_j \exp(iy_j) \quad (1 \leq j \leq 2n), \quad z_0 = \frac{t \exp(i\theta)}{z_1 \cdots z_{2n}}$$

defines a family of embeddings of this same parameterized T_0 . All absolute values remain fixed, so the family stays inside the chart, and the embeddings at $\theta = 0$ and 2π agree as parameterized maps. The induced restriction maps form a morphism from the fiber cohomology local system to the constant local system $H^k(T_0, \mathbb{R})$. Therefore, in every degree k ,

$$(5.8) \quad i_0^* M_k = i_0^* : H^k(Y, \mathbb{R}) \longrightarrow H^k(T_0, \mathbb{R}),$$

where M_k is monodromy in degree k . This statement controls the restriction of classes in degree two, as well as the torus's top-degree period. Isotopy identifies the restriction maps for T_0 and T . Thus, for $i : T \hookrightarrow Y$, $i^*(M - 1) = 0$ on $H^2(Y, \mathbb{R})$. Equations (5.4) and (5.6) give

$$(5.9) \quad i^*[\omega_K] = 0.$$

Replacing monodromy by its inverse or a positive power preserves the image in (5.4), so the conclusion is independent of this convention and of the finite base changes.

Lemma 5.1 now makes T a holomorphic Lagrangian complex torus on the rotated manifold $Y' = (Y, J)$.

5.3. The rational isotropic ray after rotation. The period plane of Y' is

$$(5.10) \quad \langle [\omega_K], [\omega_I] \rangle_{\mathbb{R}} = \langle v - a\lambda e, h \rangle_{\mathbb{R}}.$$

It is perpendicular to e , so e is an integral $(1, 1)$ class on Y' ; see Figure 1.

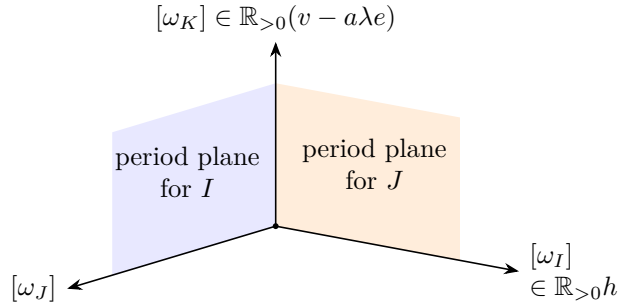


FIGURE 1. The period planes before and after rotation, inside the positive three-space spanned by the hyperkähler forms. The class e is perpendicular to $[\omega_J]$ and $[\omega_K]$, so it becomes a $(1, 1)$ class for J ; its pairing with $[\omega_J]$ fixes the positive sign.

Let $x \in H^{1,1}(Y') \cap H^2(Y', \mathbb{Q})$. Orthogonality to the period plane says

$$q(x, h) = 0, \quad q(x, v) = a\lambda q(x, e).$$

The three numbers $q(x, v)$, a , and $q(x, e)$ are rational, whereas $a \neq 0$ and $\lambda \notin \mathbb{Q}$. Therefore

$$(5.11) \quad q(x, e) = q(x, v) = 0.$$

The restriction of q to $H^{1,1}(Y', \mathbb{R})$ has signature $(1, b_2 - 3)$. In a Lorentzian space, the orthogonal complement of a nonzero null vector is negative semidefinite with radical its own line. Applying this to e and (5.11) proves

$$(5.12) \quad \begin{aligned} q(x) &\leq 0 \quad \text{for } x \in H^{1,1}(Y') \cap H^2(Y', \mathbb{Q}), \\ q(x) &= 0 \quad \implies \quad x \in \mathbb{Q}e. \end{aligned}$$

In particular, Y' is nonprojective: a projective hyperkähler manifold has an integral ample class of positive BBF square. This is the necessary direction of the projectivity criterion [21, 22]. The rational Picard space may contain negative-square classes; its isotropic classes are exactly those on $\mathbb{Q}e$. Equation (5.6) places e in the closure of the positive cone containing the Kähler class $[\omega_J]$.

5.4. A semiample point and deformation back to X . We state explicitly the two external results used in this step.

Theorem 5.2 (Greb–Lehn–Rollenske [15, Theorem 4.1]). *If a nonprojective compact irreducible hyperkähler manifold Z of complex dimension $2n$ contains a holomorphic Lagrangian complex torus S , then its algebraic dimension is n and it has a holomorphic algebraic reduction $Z \rightarrow B$ which is a Lagrangian fibration with S as a fiber. In particular, this fibration has normal projective base and provides a nontrivial semiample pullback bundle on Z .*

The projectivity assertion follows from the line-bundle construction of the algebraic reduction in [15, Section 4.2] and Stein factorization.

The fixed-class deformation theorem. Let N be a compact simply connected differentiable manifold of hyperkähler type with maximal holonomy, let $\mathcal{T}^\circ(N)$ be one connected component of its Teichmüller space, and fix a nonzero primitive integral class e with $q(e) = 0$. Write $\mathcal{C}_I$ for the component of the positive cone containing the Kähler cone of (N, I) , and set

$$\mathcal{T}_e^\circ = \{[I] \in \mathcal{T}^\circ(N) : e \in H^{1,1}(N, I), e \in \overline{\mathcal{C}_I}\}.$$

This is the positive-sign locus of [40, Definition 2.6]. By [40, Proposition 2.7] it is connected. Let E_I denote the unique holomorphic bundle with class e . Define $\mathcal{T}_{e,\text{nef}}^\circ$ and $\mathcal{T}_{e,\text{sa}}^\circ$ by requiring E_I to be nef and semiample, respectively.

Theorem 5.3 (Soldatenkov–Verbitsky [40, Theorem 3.7(v)]). *With this notation,*

$$(5.13) \quad \mathcal{T}_{e,\text{sa}}^\circ \neq \emptyset \implies \mathcal{T}_{e,\text{nef}}^\circ = \mathcal{T}_{e,\text{sa}}^\circ.$$

Its family formulation, [40, Theorem 3.8], says that a nef isotropic bundle is semiample when its pair deforms, in a smooth connected family carrying a line bundle, to a semiample pair. We use (5.13) directly.

Apply Theorem 5.2 to (Y', T) . Choose an ample bundle H on its fibration base and let A be its pullback to Y' . The bundle A is semiample and nef, and its first Chern class b is nonzero: a semipositive representative of a generated power has positive trace on an open set where the map has positive rank. Since the base has dimension n , $\int_{Y'} b^{2n} = 0$. Fujiki's relation [13] gives $q(b) = 0$. Equation (5.12) and the primitivity of e now give

$$b = ke, \quad k \in \mathbb{Z}.$$

Both b and e pair positively with $[\omega_J]$: for b this is the nonzero nef null-vector property, and for e it is (5.6). Hence $k > 0$.

Let $E_{Y'}$ be the unique bundle on Y' with first Chern class e . Injectivity of the first Chern class map gives

$$A \simeq E_{Y'}^{\otimes k}.$$

A globally generated positive power of A is therefore a globally generated positive power of $E_{Y'}$. Thus the *specified primitive class* e is semiample on Y' .

The marking furnished by Proposition 4.1 identifies Y with a point of the same ambient component $\mathcal{T}^\circ(N)$ as the original X . The hyperkähler twistor sphere connects I to J on that same marked differentiable manifold, so Y' also belongs to this component. On X , e is of type $(1, 1)$ and has the positive sign by (5.1). On Y' the corresponding statements follow from (5.10) and (5.6). Consequently both endpoints lie in the same positive fixed-class locus $\mathcal{T}_e^\circ$. The semiample point Y' makes its semiample locus nonempty; the original point X belongs to its nef locus by hypothesis. Equation (5.13) now proves that E_X is semiample on X .

Choose $s > 0$ such that $E_X^{\otimes s}$ is globally generated. Then $L^{\otimes s} \simeq (E_X^{\otimes s})^{\otimes m_0}$ is globally generated as well. In particular, on the original manifold the evaluation morphism

$$H^0(X, L^{\otimes s}) \otimes_{\mathbb{C}} \mathcal{O}_X \longrightarrow L^{\otimes s}$$

is surjective everywhere. This proves the semiampleness assertion of Theorem 1.1.

5.5. The fibration defined by a generated power. We derive the fibration description from semiampleness, using the Fujiki argument for Lagrangian fibers of Matsushita [30, 32]. Let $m > 0$ make $L^{\otimes m}$ globally generated, let

$$g : X \longrightarrow Z \subset \mathbb{P}H^0(X, L^{\otimes m})^*$$

be its morphism to the image, and take its Stein factorization $g = \nu \circ f$:

$$X \xrightarrow{f} B \xrightarrow{\nu} Z.$$

The space B is normal, f has connected fibers, and ν is finite. Since Z is projective, B is projective and $H := \nu^* \mathcal{O}_Z(1)$ is ample. By construction

$$(5.14) \quad L^{\otimes m} \simeq f^* H.$$

Put $\ell = c_1(L)$ and choose a Kähler class κ on X . The positive sign gives $q(\ell, \kappa) > 0$. Fujiki's formula and $q(\ell) = 0$ give the polynomial identity

$$(5.15) \quad \int_X (\kappa + t\ell)^{2n} = c_X (q(\kappa) + 2tq(\kappa, \ell))^n, \quad c_X > 0.$$

Comparison of coefficients shows

$$\int_X \ell^k \kappa^{2n-k} \begin{cases} > 0, & 0 \leq k \leq n, \\ = 0, & n < k \leq 2n. \end{cases}$$

On the other hand, a pullback of the Fubini–Study form by g represents $m\ell$, is semipositive, and has generic rank $\dim Z = \dim B$. Its k th power wedged with a Kähler form has positive integral precisely for $k \leq \dim B$. Thus

$$(5.16) \quad \dim_{\mathbb{C}} B = n.$$

Let F be a general smooth fiber. Its Poincaré dual class is a positive multiple of ℓ^n . Indeed, choose a very ample power $H^{\otimes a_0}$ and intersect n general members of its linear system on B . Their intersection is a nonempty finite set of transverse points in the smooth regular-value locus of f . Pulling back gives the union of the corresponding smooth fibers and represents $(a_0 m)^n \ell^n$. These fibers have the same homology class, by smooth proper transport along the connected regular-value locus. This proves the claim.

For $n \geq 2$, polarization of Fujiki's relation gives

$$(5.17) \quad \int_X \ell^n \kappa^{n-2} \sigma \bar{\sigma} = 0.$$

To see the vanishing directly, each summand in the polarized formula is a product of pairings of the $2n$ degree-two factors. Every copy of ℓ is orthogonal to ℓ , σ , and $\bar{\sigma}$, so it would have to pair

with a copy of κ . There are n copies of ℓ and only $n - 2$ copies of κ , and therefore every summand is zero. Using the fiber class, (5.17) gives

$$\int_F \sigma|_F \wedge \overline{\sigma|_F} \wedge \kappa|_F^{n-2} = 0.$$

In unitary coordinates this integrand is a fixed positive constant times the squared pointwise norm of $\sigma|_F$ times the Kähler volume form. Hence $\sigma|_F = 0$. For $n = 1$, the restriction of a holomorphic two-form to the curve F vanishes automatically. The fibers in question have complex dimension n by (5.16), so they are Lagrangian.

The same assertion holds for every irreducible component of every fiber by Matsushita's equidimensionality theorem [31, Theorem 1]. Its hypotheses are exactly those now established: X is Kähler, f is proper and surjective with normal base, and a general smooth fiber is Lagrangian. The holomorphic symplectic form is closed, as every holomorphic form on a compact Kähler manifold is. Thus every fiber component has dimension n and is isotropic on its smooth locus. This also gives the Lagrangian fibration used in Section 6.

Conversely, an ample bundle has a globally generated positive power, so (5.14) implies semi-ampleness of L . This proves the equivalence between semi-ampleness and the fibration description in Theorem 1.1.

5.6. Projective-space bases. We finish with the additional conclusion for projective X in Corollary 1.3.

Proof. Choose a very ample line bundle H on X , independently of L , with embedding $i : X \hookrightarrow \mathbb{P}^N$; the closed graph embedding $(f, i) : X \hookrightarrow B \times \mathbb{P}^N$ exhibits H as relatively very ample for f , so f is projective. The remaining hypotheses are supplied by Theorem 1.1; hence the companion's Theorem 1.1 [36] identifies B with $\mathbb{P}_{\mathbb{C}}^n$, and the following no-multiple-fiber paragraph there gives the divisor assertion. $\square$

6. FINITENESS OF SMOOTH DEFORMATION TYPES

We now deduce Corollary 1.2. The reduction from an arbitrary manifold with $b_2 \geq 5$ to one with a nef isotropic class is the argument of Sawon [37, Proposition 4.3]; we give it here to show precisely where the second-Betti-number hypothesis enters. Here deformation equivalence is generated by smooth proper families of compact irreducible holomorphic symplectic Kähler manifolds over connected complex bases. The dimension and second Betti number are constant in such a family.

We use the following smooth consequence of Engel–Filipazzi–Greer–Mauri–Svaldi [11, Theorem B and Corollary A.3]: in fixed dimension, compact irreducible holomorphic symplectic Kähler manifolds admitting a Lagrangian fibration have only finitely many smooth deformation types. Indeed, their Corollary A.3 deforms such a fibration to a projective one, and Theorem B gives finiteness for the projective representatives. The deformations in these results are locally trivial, meaning that each point has an analytic neighborhood which deforms as a product. They therefore preserve smoothness and give ordinary smooth deformations for the manifolds considered here.

Proof of Corollary 1.2. Let X have complex dimension $2n$ and $r = b_2(X) \geq 5$. Its rational BBF form is nondegenerate and indefinite, of signature $(3, r - 3)$. Meyer's theorem says that an indefinite rational quadratic form in at least five variables represents zero nontrivially [38, Chapter IV, §3.2, Corollary 2]. Clearing denominators and dividing by the positive divisibility supplies a primitive integral vector

$$0 \neq e \in H^2(X, \mathbb{Z}), \quad q(e) = 0.$$

Consider periods perpendicular to e in the marked deformation component of X . The quotient $e^\perp / \mathbb{R}e$ has signature $(2, r - 4)$, so this period locus is nonempty. A generic period in it is perpendicular to no integral vector outside $\mathbb{Q}e$: each such additional orthogonality condition is a

proper hyperplane section, and there are countably many of them. Period surjectivity therefore gives a deformation X_0 of X with

$$\mathrm{NS}(X_0) = \mathbb{Z}e.$$

There are no negative integral $(1, 1)$ classes on X_0 . The Kähler-cone description recalled in Section 4 consequently identifies its Kähler cone with its positive cone. One of $e, -e$ lies in the closure of that cone; the corresponding unique line bundle is nonzero, nef, and isotropic.

Theorem 1.1 makes this bundle semiample and gives a holomorphic Lagrangian fibration on X_0 . Thus every smooth deformation type under consideration contains a manifold with such a fibration. The smooth consequence of [11, Theorem B and Corollary A.3] stated above leaves only finitely many possibilities in dimension $2n$. $\square$

The restriction $b_2 \geq 5$ enters only through Meyer’s theorem. Theorem 1.1 itself has no such restriction, since its isotropic class is part of the hypothesis.

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